SIEMENS

COMOS

Lifecycle
3D Integration Operation

Operating Manual

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indicates that minor personal injury can result if proper precautions are not taken.

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Publisher

1

AVEVA

PDMS™ and Everything3D™ (E3D) are software products from AVEVA and are hereinafter referred to simply as PDMS or E3D.

Introduction

2

Aim and delimitation of the contents

The functions and tools for 3D Integration encompasses

- Data exchange between COMOS and PDMS or E3D via the plugin "PDMS/E3D Engineering Interface". See chapter PDMS/E3D Engineering Interface (Page 11).
- Data exchange over the "COMOS-PDMS-Interface". See chapter COMOS PDMS interface (Page 65).
- Importing of pipe specs from PDMS/E3D via the plugin "PDMS/E3D Pipe Specs Import". See chapter Importing technical specifications of PDMS (Page 105).
- Display of 3D data via the application "COMOS Design Viewer". See chapter COMOS Design Viewer (Page 107).

Note

COMOS PDMS interface

This designation refers to the COMOS PDMS interface that was delivered and installed until COMOS 8.2. "PDMS/E3D Engineering Interface" designates the interface with a uniform user interface as of COMOS 10.1. Both interface versions are based on the same configuration.

Starting the PDMS interface

Depending on the PDMS interface used and the respective AVEVA software, the following command lines are intended for starting:

	PDMS	E3D	Catalogue 1.1	Catalogue 2.1
COMOS-PDMS-Interface	!!ITSetupCOMOS()	!!ITSetupCOMOS()	!!ITSetupCOMOS()	!!SAGLaunchInterface()
PDMS/E3D Engineering Interface	!!SAGLaunchInterface()	!!SAGLaunchInterface()	!!SAGLaunchInterface()	!!SAGLaunchInterface()

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3.1 Introduction

Objective

With the PDMS/E3D Engineering Interface you as a COMOS user synchronize 2D engineering data from COMOS and 3D data from PDMS or E3D in real time in online mode. In the process you exchange data in both directions between the target and source system.

The PDMS/E3D Engineering Interface, which is abbreviated as Interface below, is a component that you call up respectively in both systems and whose basic principle is identical in both systems.

Basic principle and functional principle of the Interface

You transfer and synchronize data in accordance with this principle:

Action	Description
Requesting data	In your target system you request data from the source system. This data is available during the runtime in a buffer memory in the Interface and is displayed there in the "2D workspace" or "3D workspace" control group.
Interacting	With the functions and features of the interface you compare the requested data shown in the respective workspace with the dataset in the target system.
Ascertaining	You ascertain whether inconsistencies exist between the data from the source and target system and if so, of what type these are. To this purpose each inconsistency is assigned to a status class. If there is no inconsistency, the data and the associated object is classified as "synchronized".
Resolving	You resolve determined and classified inconsistencies by means of corresponding functions. In the process, you change and synchronize the dataset in one or both databases.

You use and repeat this principle until the 2D and 3D data are consistent, error-free and synchronized. See also chapter Overview of the workflow (Page 18).

Further functionalities of the Interface

- Importing documentation to COMOS. See chapter Importing DocLinks (Page 48).
- Importing material data to COMOS. See chapter Creating material (MTO) from PDMS/E3D (Page 54).
- Navigating between objects and documents in PDMS/E3D and COMOS. See chapter Navigating (Page 58).

Documentation

If not explicitly stated otherwise, the following documentation describes the functions, features and operation of the Interface in COMOS.

This then applies correspondingly for the Interface in PDMS/E3D.

If statements apply only for one system, corresponding information or a separate chapter is provided^.

Terminology and user interface language

The user interface in PDMS/E3D is only available in English.

However, the functions and features described in the documentation can almost always be easily translated.

You will find corresponding notes when

- it is not easily possible to make sense of the meaning.
- the COMOS user interface contains English terminology that cannot be translated.
- terminology from PDMS/E3D is quoted.

3.2 Basic knowledge

3.2.1 Definition of objects

When data and objects are exchanged over the interface and are also edited in the systems, it is necessary to define and distinguish their respective status.

The distinction is made with the following:

- Object categories
- Object classes
- Status classes See also chapter Definitions for status classes and inconsistencies (Page 14).

Separations and classifications

- Are provided for users in the form of tooltips, symbols and texts in the user interface.
- Are the basics for editing with the software for commands such as "Check status" or "Resolve".

Object categories

Object category	Description
"One-side object"	Objects that do not have a counterpart in PDMS/E3D or COMOS.
"Two-side object"	Objects that are known in both systems and whose subclass contains the object type of the PDMS/E3D element.
Reference	Objects that are defined in the subclass as a reference to a "Two-side object".

Object classes

Object class	Description and color coding	Associated object category
Incorrect path	Error during evaluation of the path and name of the COMOS object in COMOS. Color: Red	"Two-side object"
Undefined base object	The COMOS base object is not defined. Color: Red	"Two-side object"
Fully defined	Color: Black	"Two-side object"
Structure engineering object	Structure objects from COMOS that are defined through name mapping and do not have a counterpart in PDMS/E3D. Color: Blue	"One-side object"
Process Line	Objects that are defined by PDMS Area Lines whose behavior, in turn, depends on the assigned subclass. Color: Blue	"One-side object"
Structure 3D object	Structure objects from PDMS/E3D that do not have a counterpart in COMOS. These objects are never generated in COMOS but can contain subobjects that are to be captured by the interface. Color: Blue	"One-side object"
Reference	Color: Blue	Reference

Additional definitions

These terms and definitions are relevant for configuring the subclasses that the administrator can use, for example, to control the display of the object tree in the respective workspace.

See also chapters COMOS: Object tree in the 3D workspace (Page 24) and PDMS/E3D: Object tree in the 2D workspace (Page 25).

Term	Description
Top-Level object	Structuring objects that are assigned in PDMS/E3D below the ZONE node. The administrator makes the specific definition using the corresponding subclass.
Sub-Level object	Objects that are located below a Top-Levelobject and that describe technical devices and components. The administrator makes the specific definition using the corresponding subclass.
Unlabeled object	3D objects such as reducers or steel construction that never have a name or label in PDMS/3ED. These objects can be captured by the interface with the corresponding configuration and are then treated just like any other object.

3.2.2 Definitions for status classes and inconsistencies

Status classes are defined for objects with object category "Two-side object" and with "Fully defined"; these status classes describe the consistency between the data in COMOS and PDMS/E3D in more detail. See also chapter Definition of objects (Page 12).

An object can have only one status class.

You determine the status class of an object by filtering, marking or with a status check. The status class for an object is displayed by an additional status symbol at the object. In addition, you can save this information in the form of a report as a file.

These methods and functionalities are available in both COMOS and PDMS/E3D. See also chapter Overview of the workflow (Page 18).

Setting status values in COMOS

In PDMS/E3D you can write the determined status class as a status value to the corresponding COMOS object in COMOS. This functionality is only available in PDMS/E3D. If the status display is activated in the COMOS Navigator, these objects and the associated P&ID are marked in color.

Status classes

The table shows the six possible status classes and the descriptions as they are shown in the user interface and in the reports:

Status class	Title bar in the report	Status value in COMOS
What's new	Missing objects in COMOS or PDMS/E3D	1
Attributes not synchronized	Objects not synchronized objects at attribute level	3
Tagged differently	Objects with different tags in COMOS and PDMS	2
Not assigned	Existing objects that are not assigned	5

Status class	Title bar in the report	Status value in COMOS
Improperly assigned	Objects with incorrect assignment information	6
Synchronized	Objects synchronized at object level and attribute level	4

Note

Additional status value in COMOS

In COMOS the value 7 is assigned additionally to the status values listed in the table. It describes and identifies inconsistent child objects.

3.2.3 Status check

You use the following button in the "Filtering and marking" area for the status check and for calling additional functionalities; it can be activated and deactivated:

If the button is activated:

- The requested objects are marked in the workspace with status symbols in accordance with the status classes. See also chapter Definitions for status classes and inconsistencies (Page 14).
- Additional commands can be called from the selection menu by clicking the arrow symbol.

If you deactivate the button, the status symbols are hidden again and the commands are deactivated in the selection menu.

The functionalities are available in COMOS and PDMS/E3D, with the exception stated in the table. See also chapter Overview of the workflow (Page 18).

Commands in the selection menu

Command	Description
"Refresh"	Updates the displayed information in the sense of a refresh by reading in the COMOS or PDMS/E3D data again.
"Report"	Creates a report in the form of a file with all objects and their status descriptions. See also chapter Generating reports (Page 37).
"Set status in COMOS"	Only available in PDMS/E3D: Assigns a status value for each possible status class and writes it to the COMOS object in COMOS. See also chapter Definitions for status classes and inconsistencies (Page 14).

3.2.4 Filtering and marking

You use the following buttons in the "Filtering and marking" area for filtering and marking and for calling additional functionalities:

Activated tab	Display of the button
"Filter"	
"Mark "	

They have the following features:

- Depending on the tab selected, the display and the functionality of the button changes.
- The button activates and deactivates the filtering or marking condition that you have selected in the standard table.
- If the button is activated and you select a different filter or marking condition, the condition is applied immediately to the data in the workspace.
- If the button is activated, additional commands can be called from the selection menu by clicking the arrow symbol.

The functionalities are available in COMOS and PDMS/E3D.

Commands in the selection menu

Command	Description
"Refresh"	Updates the displayed information in the sense of a refresh by reading in the data again.
"Resolve"	Starts the respective operation for resolving the inconsistency for those objects that fulfill the filtering or marking condition. See also chapter Resolving inconsistencies (Page 16).
"Report"	Creates a report in form of a file with those objects that fulfill the filtering and marking condition. See also chapter Generating reports (Page 37).

3.2.5 Resolving inconsistencies

An operation for resolving is provided for each status class that describes an inconsistency.

These operations are available at the following points in the user interface:

- In the toolbar of the Interface
- in the "Attributes for current selection" control group
- In the shortcut menu of an object in the 2D or 3D workspace
- Can be called via the "Resolve" command in the "Filter and mark" control group. See also chapter Filtering and marking (Page 16).

Operations and procedures for resolving

Status class	Operation for resolving	Can be called up via the "Resolve" command	Description and procedure
What's new	Creating	No	See chapter Creating objects in COMOS (Page 38). See chapter Creating objects under any owner (Page 41).
Attributes not synchronized	Updating attributes	Yes	See chapter Updating attributes (Page 36).
Tagged differently	Only available in PDMS/E3D: Update the names by using "Update names"	Yes	See chapter Updating names in PDMS/E3D (Page 47).
Not assigned	In PDMS/E3D: "Assign by matching names" In COMOS: "Assign"	Yes	See chapter Assigning COMOS objects and PDMS/E3D objects (Page 46).
Improperly assigned	Canceling the assignment	Yes	See chapter Canceling the assignment between COMOS objects and PDMS/E3D objects (Page 40).

3.2.6 Context menu of an object in the object tree

The objects in the object tree in the respective workspace offer a context menu with different commands in COMOS and PDMS/E3D.

See also chapters COMOS: Object tree in the 3D workspace (Page 24) and PDMS/E3D: Object tree in the 2D workspace (Page 25).

Commands that are available in the context menu of an object

In COMOS	In PDMS/E3D	Description
Navigate to PDMS/E3D	Navigate to PDMS/E3D	Navigates to the PDMS/E3D structure
Navigate to COMOS	Navigate to COMOS	Navigates to the COMOS object tree in the Navigator
--	Navigate to COMOS P&ID	Navigates to the COMOS document in the Navigator
Zoom in 3D	--	Zooms in on object in PDMS/E3D in the 3D model
Navigate to the reference	--	Navigates to the reference if one exists
--	Assign to selected	Assigns the object in the object tree at which the command is called to the previously selected PDMS/E3D object
--	Assign by matching names	Assigns objects based on matching names
Canceling the assignment	Unassign	Cancels the assignment between COMOS and PDMS/E3D objects

In COMOS	In PDMS/E3D	Description
Check status	Check status	Checks the status and assigns a status class
Update attributes	Update attributes	Updates all attributes of the object
Update attributes (recursively)	Update attributes (recursively)	Updates the attributes of the object and its child elements
--	Update names	Updates names
Create	Create	Creates a new object
--	Create (beneath selected)	Creates a new object below the object that was selected in PDSM/E3D
Create hierarchy	Create hierarchy	Creates a hierarchical structure
--	Create hierarchy (beneath selected)	Creates a new object with associated structure below the object that was selected in PDSM/E3D
--	Show potential owners	Shows all potential owners of the object
Import MTO	--	Imports material data to COMOS

3.3 Overview of the workflow

Requirement

You are familiar with the basic principle of the Interface. See also chapter Introduction (Page 11).

Basic workflow

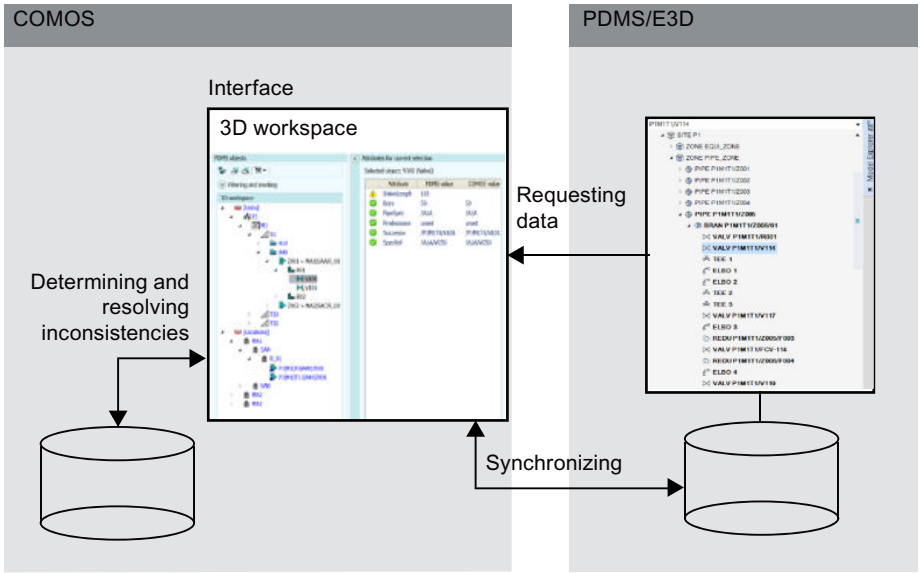
1. In both programs start the Interface by using the "PDMS/E3D Engineering Interface" plugin respectively.
   See chapter Opening the COMOS PDMS/E3D Engineering Interface (Page 27).
2. Depending on the task request the data of the source system in the target system.
   As a result the required data are displayed in the respective workspace as an object tree. These data represent structuring objects such as folders as well as technical engineering objects such as pumps, valves or pipes.
   See Chapter Requesting COMOS objects (Page 29) and Requesting PDMS/E3D objects (Page 28).

3. Determine inconsistencies.
   The following functionalities are foreseen for determining:
   - Quick check by selecting an object
   - Check status
   - Filtering
   - Marking

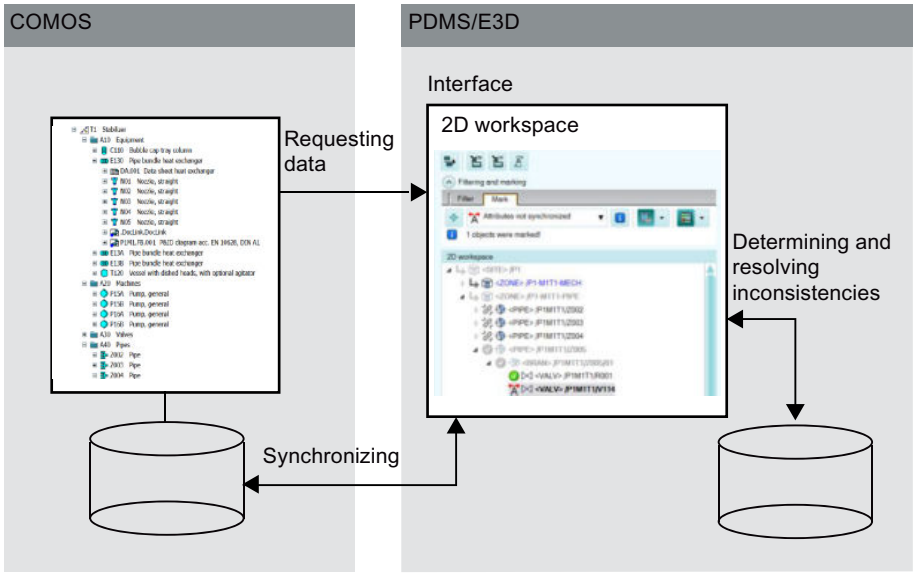
   See chapter Determining inconsistencies (Page 30).
4. Resolving inconsistencies and synchronizing data.
   In the process either all, individual or filtered objects can be edited.
   The following operations are foreseen:
   - Update attributes and names
   - Creating
   - Canceling the assignment
   - Assigning
     See Chapter Resolving inconsistencies (Page 35) and Resolving inconsistencies (Page 16).
   - If required, create reports in file form that list the objects and their status classes in tables. See chapter Generating reports (Page 37).

The fundamental workflow is identical in COMOS and PDMS/E3D.

Overview: Working with the Interface in COMOS:

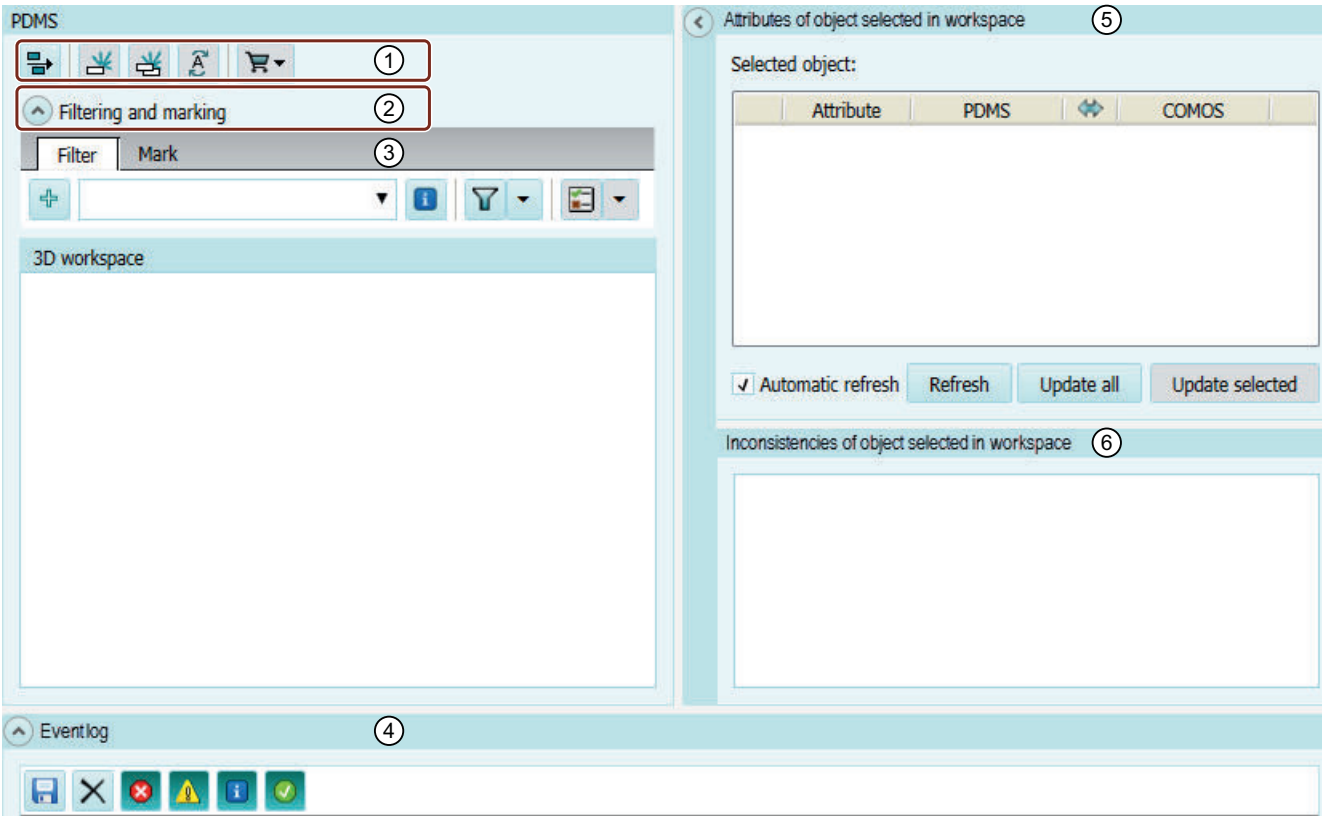


Overview: Working with the Interface in PDMS/E3D:



3.4 User interface description

3.4.1 COMOS: User interface description



"PDMS" area

Number	Description
1	• "Load workspace" button Load PDMS/E3D objects in the "3D workspace" area. • "Create all" and "Create hierarchy for all" buttons: See chapter Creating objects in COMOS (Page 38). • "Update attributes for all" button Updates the attribute values for all objects in the "3D workspace" area. • "Import material for all" button Menu commands: – Current Imports material labeled as current for all pipes of the PDMS type "PIPE" in the 3D workspace. – Released Imports material labeled as released for all pipes of the PDMS type "PIPE" in the 3D workspace.
2	"Filtering and marking" control group • "Filter" and "Mark" tabs The respective tab defines whether objects are filtered or marked. • Standard table Contains the status classes by which objects can be filtered or marked in the "3D workspace" area. See also chapter Definitions for status classes and inconsistencies (Page 14). • Buttons – Shows the status classes with descriptions and status symbols as information. Behaves like a tooltip, no clicking required. – Depending on the selected tab, activating and deactivating of filtering or marking function. – Check status
3	"3D workspace" control group Displays the requested PDMS/E3D objects as an object tree.

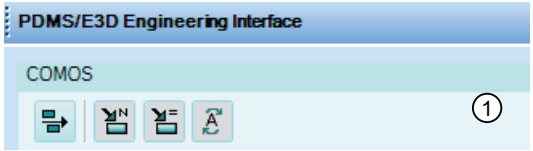
"Event log" area

Number	Description
4	Shows the status of the actions that you have started.

"Attributes of object selected in workspace" area

Number	Description
5	Shows the attributes of the object that is selected in the "3D workspace" area. Columns • "Attribute" Name of the attribute • "COMOS" Attribute value from COMOS • Double-arrow symbol If an arrow symbol is displayed in a line, this indicates differing attribute values. If you want to synchronize the values, click the arrow symbol. The direction of the arrow indicates the direction of the transfer. • "PDMS" Attribute value from PDMS Buttons • "Update selected" If you have marked several rows with differing attribute values and want to synchronize these, click this button. • "Update all" If you want to synchronize all lines with differing attribute values, click this button. Preceding marking of the rows is not required. • "Reevaluate" Updates the displayed information in the sense of a refresh by reevaluating the COMOS data. "Evaluate automatically" option • Activated: Automatically executes the functionality of the "Reevaluate" button.
6	"Inconsistencies of object selected in workspace" control group Shows the inconsistencies between the COMOS objects and the PDMS/E3D objects. The inconsistencies are determined recursively and displayed accordingly.

3.4.2 PDMS/E3D: User interface description



The English user interface of the Interface in PDMS/E3D is identical in content to that in COMOS with the exception specified.

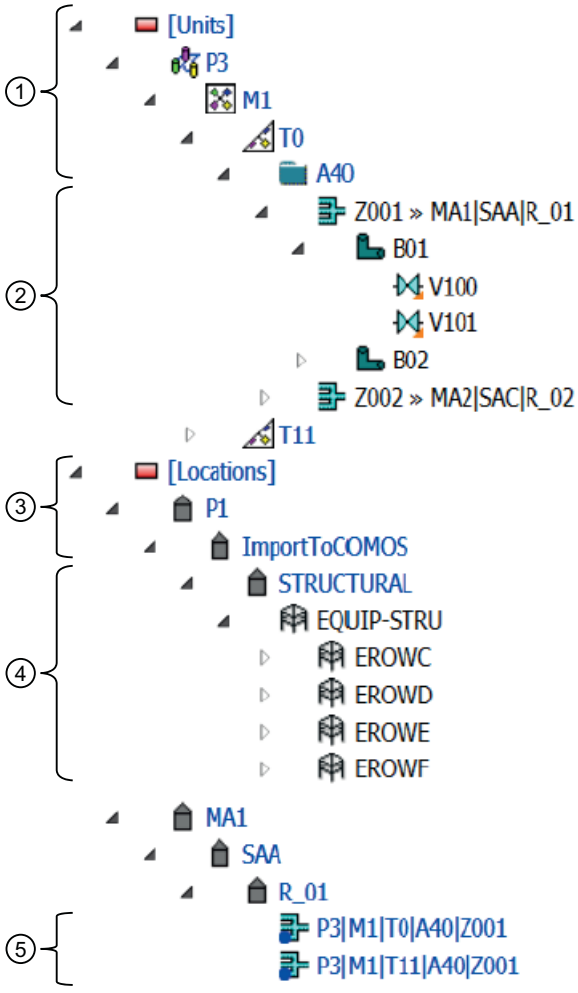
"COMOS" area

Number	Description
1	• "Assign all by matching names" button Assigns objects from COMOS and PDMS/E3D recursively on the basis of correlative names. See also Assigning COMOS objects and PDMS/E3D objects (Page 46). • "Update all names" button Updates the name in PDMS/E3D. See also chapter Updating names in PDMS/E3D (Page 47).

3.4.3 COMOS: Object tree in the 3D workspace

When data from PDMS/E3D are requested, these are displayed as object tree in the 3D workspace in COMOS. See also chapter Requesting PDMS/E3D objects (Page 28).

Object tree in the 3D workspace in COMOS



① Expected unit structure in COMOS that the administrator has specified in the subclass definition.

② Objects that are placed in PDMS/E3D below "ZONE".

③ Expected location structure in COMOS that the administrator has specified in the subclass definition.

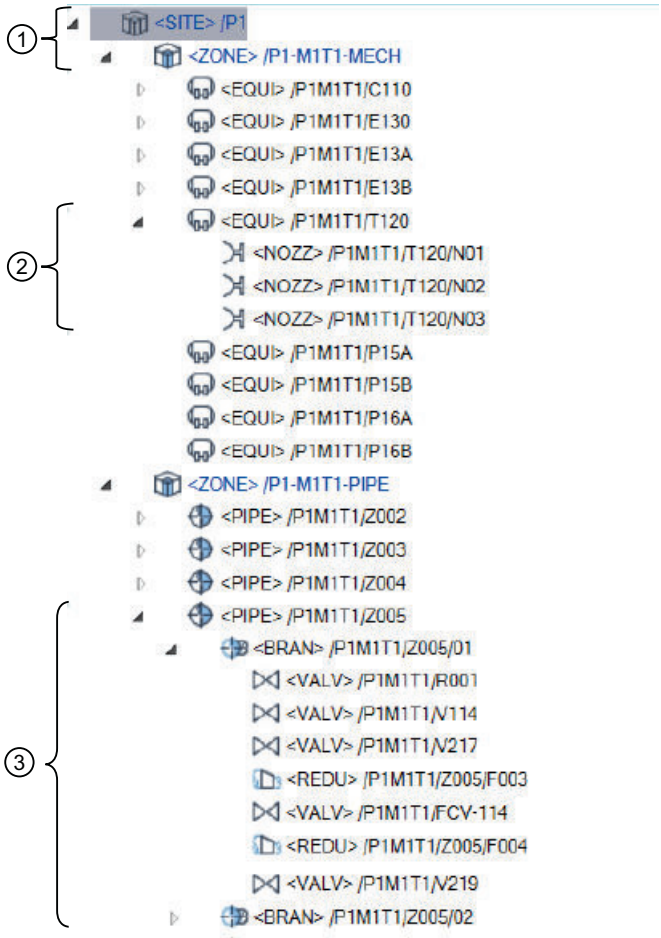
④ Objects that are placed in PDMS/E3D below "ZONE".

⑤ References to unit or location structure. Identified by blue icon fading.

3.4.4 PDMS/E3D: Object tree in the 2D workspace

When data from COMOS are requested, these are displayed as object tree in the the 2D workspace in PDMS/E3D. See also chapter Requesting COMOS objects (Page 29).

Object tree in the 2D workspace in PDMS/E3D



① Grouping by sites and zones according to the convention in PDMS/E3D and configuration of the administrator. Terminology in PDMS/E3D: "SITE" and "ZONE".

② Hierarchical display of COMOS objects, such as machines or equipment

③ Hierarchical display of pipes from COMOS with branches and their components, such as valves. Terminology in PDMS/E3D: "PIPE" and "BRANCH".

3.5 Requirements for the PDMS/E3D Engineering Interface

3.5.1 General requirements

A COMOS Administrator creates and verifies the following requirements:

- The software requirements are met.
- PDMS/E3D Engineering Interface has been installed and activated.
- The COMOS PDMS/E3D Engineering Interface has been entered in PDMS or E3D as additional program component.

- An administrator has defined which COMOS objects may be loaded for the respective user name or the user group.
- An Administrator has synchronized the classes and subclasses in COMOS and PDMS. Synchronizing is not required:
  - When the Interface is used for the first time.
  - When the settings of the "Folders for class/subclass definitions" object have been changed.
  - When a new class or subclass definition object has been created.
  - When the settings of a subclass definition object have been changed.

You can find additional information on this topic in the "3D Integration Administration" manual.

3.6 Working with the PDMS/E3D Engineering Interface

3.6.1 Opening the COMOS PDMS/E3D Engineering Interface

In order to use the interface, it must be open in COMOS and in PDMS or E3D.

Note
Working in network mode

If you work in network mode, you first have to start the interface in COMOS.

Note
Plug-in in COMOS available with limited functionality under certain conditions

Only the "Event log" area appears in COMOS in the following cases.

- If there is no connection to PDMS/E3D
- When you open the PDMS Paragon plug-in.

The following functionalities are not available.

Requirement

- COMOS and PDMS/E3D are open.
- The general requirements are met. See also chapter General requirements (Page 26).

Procedure

1. In COMOS, select the menu "Plugins > CAE Apps > AVEVA > PDMS/E3D Engineering Interface".
2. Switch to PDMS/E3D.
3. Call the `!!SAGLaunchInterface()` function in the command line.

Result

The programs are interconnected by means of the interface. The user interface of the interface opens in COMOS and in PDMS/E3D.

- See also chapter Overview of the workflow (Page 18).
- See also chapter PDMS/E3D: User interface description (Page 23).
- See also chapter COMOS: User interface description (Page 21)

3.6.2 Requesting PDMS/E3D objects

To synchronize data in COMOS with the 3D model, request objects from PDMS or E3D.

Objects can be created in a unit structure or a location structure. An administrator can configure this behavior. You can find additional information on this topic in the "3D Integration Administration" manual, keyword "Defining structural behavior".

Requirement

- The subclass definition objects are configured. See also chapter General requirements (Page 26).
- The Interface is open. See also chapter Opening the COMOS PDMS/E3D Engineering Interface (Page 27).

Procedure

1. Click the "Request data" button on the toolbar:
2. Select one or multiple subclasses in the "Request data from PDMS/E3D" window. Depending on which subclasses you select, the object types contained therein are eligible for import into the interface.
3. In PDMS/E3D select a node whose child nodes are to be imported into the interface.
4. Click "OK".

Result

An object tree containing the imported objects is displayed in the "3D workspace" area. See also chapter COMOS: Object tree in the 3D workspace (Page 24).

3.6.3 Requesting COMOS objects

To synchronize data in PDMS/E3D with the 2D model, request objects from COMOS.

Requirement

- An administrator has defined which COMOS objects may be loaded for your user name or your user group. See also chapter General requirements (Page 26).
- The interface is open. See also chapter Opening the COMOS PDMS/E3D Engineering Interface (Page 27).

Procedure

1. Click the "Request data" in the toolbar. A dialog window opens.
2. Select the designation of the desired workspace from the drop-down list.
3. Click the "Request" button to confirm the selection.

Result

The objects are displayed hierarchically in an object tree in the workspace. See also chapter PDMS/E3D: Object tree in the 2D workspace (Page 25).

- Black: These objects exist in COMOS and in PDMS/E3D.
- Blue: These objects only exist in one of the two applications.

If you have requested a two-level pipe structure conforming to KKS which consists only of one pipe section and the pipe parts below it, the following is created by default in the Interface:

Example:

COMOS (KKS)	PDMS/E3D Engineering Interface
Pipe section	Pipe
	Virtual pipe section
Valve	Valve

3.6.4 Determining inconsistencies

Various methods are available for determining inconsistencies for the objects in the "2D workspace" or "3D workspace":

- Check status
  - You can check all objects or only one object. This results in each object having a status icon in accordance with the status class assigned to it.
  - See chapter
    Checking the status for all objects (Page 30)
    Checking the status for individual objects (Page 31)
    Checking the status of the attributes of an object (Page 31)
- Filtering or marking
  - You can filter or mark the objects in the object tree in accordance with the status classes.
  - See chapters Filtering objects (Page 32) and Marking objects (Page 33).
  - In addition to the predefined status classes, you can also use customized conditions.
  - See chapter Creating user-defined criteria (Page 34).

3.6.4.1 Checking the status for all objects

You check the status for all objects in the workspace by using the corresponding command in the "Filtering and marking" area.

Requirement

- You are familiar with status classes, inconsistencies and status check. See also chapters Definitions for status classes and inconsistencies (Page 14) and Status check (Page 15).
- The workspace contains objects.

Procedure

1. Expand the "Filtering and marking" section.
2. Click the "Check status" button.

Result

- The button is activated.
- The commands in the selection menu of the button are activated so that the creation of reports is possible.
- The objects have status icons in accordance with the status classes.

Deactivating the status check

Click the button again to deactivate the status check and hide the status symbols at the same time.

3.6.4.2 Checking the status for individual objects

Requirement

- The context menu of an object in the object tree is known. See also chapters COMOS: Object tree in the 3D workspace (Page 24) and PDMS/E3D: Object tree in the 2D workspace (Page 25).
- The workspace contains objects.

Procedure

1. Select an object in the object tree.
2. Select the "Check status" command in the context menu of the object.

Result

- A status symbol in accordance with the status classes is shown at all objects recursively. See also chapter Definitions for status classes and inconsistencies (Page 14).
- The "Check status" button in the "Filtering and marking" area is activated.

3.6.4.3 Checking the status of the attributes of an object

Requirement

- The workspace contains objects.
- The COMOS object and PDMS/E3D object are assigned to each other.
- An administrator has specified which attributes are checked for consistency.

Procedure

1. Select an assigned object in the object tree.
2. "Attributes of object selected in workspace" area
   - Case 1: A message is shown if an object is not correctly connected or does not contain any attributes to be verified.
   - Case 2: The attributes are shown if the object is correctly connected.
3. For Case 2: In order to ensure that the displayed data correspond to the current database data click the "Reevaluate" button.

Result

A symbol is shown at all attributes which have been checked for consistency.

- Green circle: Attribute value in COMOS matches PDMS/E3D.
- Yellow triangle: Attribute value in COMOS does not match PDMS/E3D.

Eliminating inconsistent attribute values

Use the functions and buttons for updating in the "Attributes of object selected in workspace" area.

See also chapters COMOS: User interface description (Page 21) and Updating attributes (Page 36).

3.6.4.4 Filtering objects

Requirement

- You have a basic knowledge of filtering and marking. See also chapter Filtering and marking (Page 16).
- The workspace contains objects.

Procedure

1. Expand the "Filtering and marking" section.
2. Select the "Filter" tab.
3. Open the standard table and select the filter condition for a status class. See also chapter Definitions for status classes and inconsistencies (Page 14).
4. Click the "Toggle filter" button.

Result

- The button is activated.
- All objects that do not match the filter are hidden along with their parent objects. The other objects are highlighted in color.

Legend:

- Black: Objects that match the filter.
- Gray: Parent objects that do not match the filter.
- Blue: Objects that match the filter and have an undefined subclass.
- Light blue: Objects that do not match the filter and have an undefined subclass.

Note

Recursive operations are also applied to hidden objects

When you perform a recursive operation on an object after filtering, the operation is also applied to hidden child objects which do not match the filter conditions.

Canceling filtering

Click the "Toggle filter" button again.

3.6.4.5 Marking objects

Requirement

- Basic knowledge of filtering and marking. See also chapter Filtering and marking (Page 16).
- The workspace contains objects.

Procedure

1. Expand the "Filtering and marking" section.
2. Select the "Mark" tab.
3. Open the standard table and select the condition for a status class. See also chapter Definitions for status classes and inconsistencies (Page 14).
4. Click the "Toggle mark" button.

Result

- The button is activated.
- Objects that match the condition are highlighted in black bold font. The color of the other objects does not change.

Removing a mark

Click the "Toggle mark" button again.

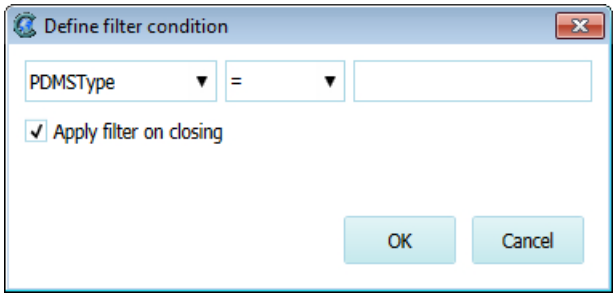
3.6.4.6 Creating user-defined criteria

You can create your own filters or conditions for a marking which filters the objects in the "PDMS objects" area according to the value of the following properties:

- PDMS type
- PDMS name
- COMOS name

Procedure

1. Expand the "Filtering and marking" section.
2. Select the corresponding tab, depending on whether you want to create a new filter or a condition for a mark.
3. Click on the "Define new condition" button.
   The "Define Filter Condition" window opens.



4. Select the property according to which filtering or marking is to take place.
5. Select an operator.
   - Operator "="
     Objects match the condition exactly.
   - Operator "Contains"
     Object contains the string that you enter as a condition.
6. Enter a condition in the right field.
7. Optional: Activate the option "Apply filter on closing".
   The new condition is applied once you click OK.
8. Click "OK".

Result

The new condition is included in the standard table of the filters. It is available in the current session.

See also chapters Filtering objects (Page 32) and Marking objects (Page 33).

3.6.5 Resolving inconsistencies

3.6.5.1 Using the "Resolve" command

Requirement

- You are familiar with the basics of resolving inconsistencies. See also chapter Resolving inconsistencies (Page 16).
- You have filtered objects in the respective workspace. See also chapter Filtering objects (Page 32).

Procedure

1. From the selection menu of the buttons for filtering, select the command "Resolve".

Result

- The objects are edited and synchronized according to the provided operations.
- The status symbols were updated.

3.6.5.2 Synchronizing individual objects

Requirement

- You are familiar with the basics of status classes and resolving inconsistencies. See also chapters Definitions for status classes and inconsistencies (Page 14) and Resolving inconsistencies (Page 16).
- The context menu of an object is known. See also chapter Context menu of an object in the object tree (Page 17).
- An object tree is displayed in the respective workspace.
- You have checked the status for an object. See also chapter Checking the status for individual objects (Page 31).

Procedure

1. Open the context menu of the object.
2. Depending on the status class of the object, select the suitable command from the context menu.
3. Optional: To ensure that the object in the workspace is always compared to the latest database from COMOS or PDSM/E3D, select "Check status" from the context menu.
4. Optional: If the status class is not "Synchronized", select a command again from the context menu.

Result

- The object is synchronized.
- The status symbol is updated.
- Optional: The color coding for the object category and object class is updated.

3.6.5.3 Updating attributes

Requirement

- You are familiar with the fundamentals of status classes and inconsistencies. See also chapter Definitions for status classes and inconsistencies (Page 14).
- The commands and buttons for updating attributes are known. See also chapter COMOS: User interface description (Page 21).
- An object in the object tree is selected.
- The selected object is assigned to an object in the other system at the attribute level.

Procedure

- To update only the attributes of the currently selected object, select the "Update attributes" command in the shortcut menu.
- To update the attributes of the currently selected object and the attributes of the objects below it, select the command "Update attributes (recursively)" in the shortcut menu.
- To update all the attributes in the "Attributes for current selection" area, select the "Update all" button.
- To update selected attributes in the "Attributes for current selection" area, select the desired attributes and click the "Update all" button.
- To update only individual attributes in the "Attributes for current selection" area, select the respectively visible "Arrow" button.

Procedure for all objects

- To update attributes for all objects in the object tree click the "Update attributes for all" button in the toolbar.

Result

The attribute values are updated according to the settings in the subclass of the COMOS object. Depending on the setting, the attribute values are transferred from PDMS/E3D to COMOS or vice versa.

- If you eliminate inconsistencies and this makes adjustments necessary in PDMS/E3D, the PDMS/E3D data is synchronized with the data already loaded in the workspace and updated.
- If you eliminate inconsistencies and adjustments in COMOS are required in the process, they are taken from the PDMS/E3D object and written to the corresponding COMOS object.

3.6.5.4 Generating reports

You generate reports with status information via the corresponding command in the selection menus that can be called up by using the buttons in the "Filtering and marking" area.

The reports contain a tabular listing of the objects and their status classes as well as additional data such as the file name or date.

The procedure is identical in COMOS and PDMS/E3D.

Requirement

- You have a basic knowledge of filters, marking and status checks. See also chapters Filtering and marking (Page 16) and Status check (Page 15).
- You have filtered or marked or checked the status of objects in the workspace.
- The corresponding button is activated.

Procedure

1. Select the "Report" command from the selection menu of the respective button.
2. A dialog window for saving the file is opened.
3. Enter a name and a storage location for the file.
   Alternatively: Leave the default entries unchanged.
4. Confirm your entries.

Result

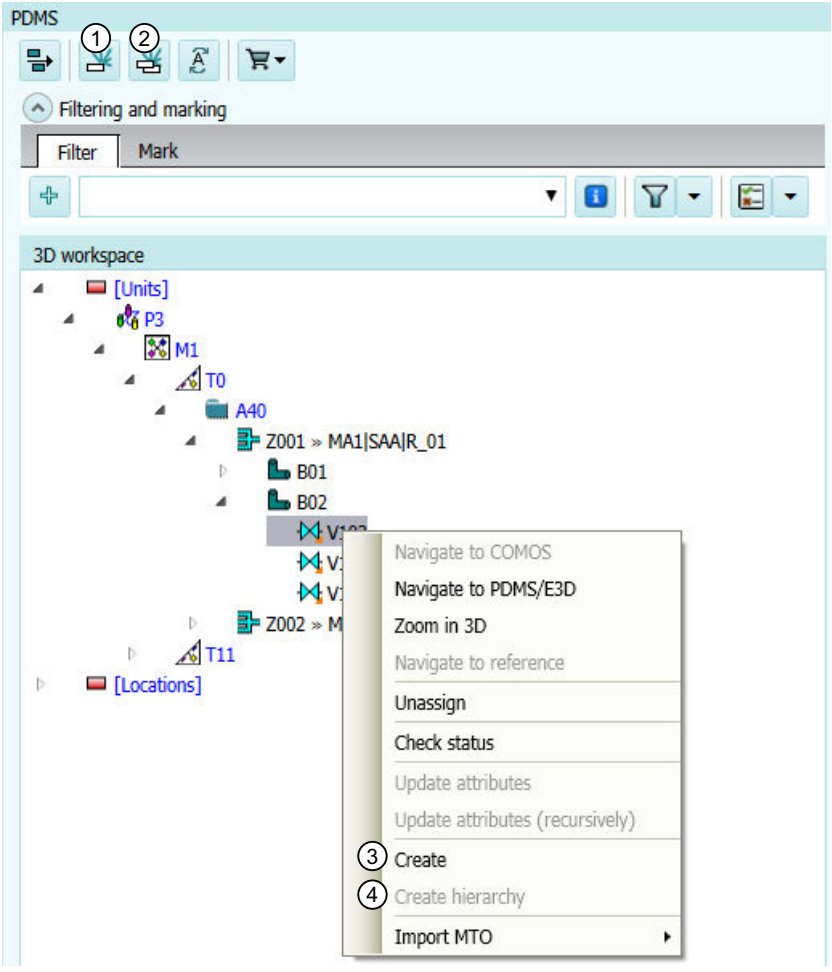
- The report is stored in form of an HTML file.
- Depending on your Windows configuration the file content is displayed immediately in your browser.

3.6.5.5 COMOS: Creating data for 2D planning

Creating objects in COMOS

You have the following options of generating objects:

- By using the context menu in the "3D workspace" you create an individual object and the associated structured objects, if these do not exist yet.
- By using the toolbar you create all objects and the associated structured objects, if these do not exist yet.



Requirement

- You have requested PDMS/E3D objects.
- Subclasses have been configured for the objects you want to create.

Creating objects using the toolbar

Select one of the following commands:

Number	Commands	Description
①	"Create all"	Creates all objects and primary structuring objects, if these are missing.
②	"Create hierarchy for all"	Creates all objects and their subordinate objects in the COMOS Navigator. Also creates primary structuring objects, if these are missing.

Creating objects using the context menu

1. Open the context menu of a PDMS/E3D object in the "3D workspace".
2. Select one of the following commands:

Note

Creation location in COMOS is different for Area Lines

The location where objects are created in COMOS when Area Lines are created is different from the location where the objects are displayed in the interface.

Number	Commands	Description
③	"Create"	The command is available at the objects that represent technical engineering objects. Creates the object and primary structuring objects, if these are missing.
④	"Create hierarchy"	Creates the object and its subordinate objects in the COMOS Navigator. Also creates primary structuring objects, if these are missing.
-	"Create hierarchies (Process Lines/Area Lines)"	Available at Process Lines in the "Units" node. Creates the following objects: • Process Line • All Area Lines that reference the Process Line • Components of the Process Line

Result

The objects are created in COMOS in the Navigator and are assigned to the related PDMS/E3D objects. If an object cannot be created, the software continues with the next object.

The "Progress" area indicates which objects could not be created.

Canceling the assignment between COMOS objects and PDMS/E3D objects

Procedure

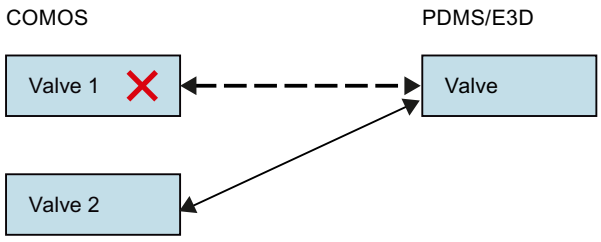
In the "3D workspace", click the "Unassign" command in the context menu of an assigned object.

Result

- For a correct/unique connection:

  The connection information is deleted recursively at the COMOS object and PDMS/E3D object.

- For an incorrect/non-unique connection:



  The connection information for the COMOS object is deleted recursively. When the PDMS/E3D object has a correct connection to another COMOS object, this connection information is retained.

When you activate the "Not assigned" filter, the objects without connection information are displayed as new objects.

See also

Connection information for labeling an assignment (Page 45)

Filtering objects (Page 32)

3.6.5.6 PDMS/E3D: Creating data for 3D planning

Creating objects under a pre-defined owner

Requirement

- The administrator has enabled and configured "Use ZONE and SITE-Mapping ".
- The "2D workspace" area contains objects from COMOS.
- The objects do not exist in PDMS/E3D.

Procedure

1. To find out which objects do not yet exist in PDMS/E3D, filter by "What's new?". See also chapter Filtering objects (Page 32).
2. In the "2D workspace" section, open the shortcut menu of the object that you want to create in PDMS/E3D.
3. Select one of the following commands:
   - "Create"
     Searches in PDMS/E3D for the object that corresponds to the owner of the selected object in COMOS. Creates the missing object under the owner found.
   - "Create hierarchy"
     Searches in PDMS/E3D for the object that corresponds to the owner of the selected object. Creates the missing object and all underlying objects under the owner found. If an object cannot be created, the software continues with the next object.

Result

The objects are created in PDMS/E3D and are assigned to the related COMOS objects.

See also

Connection information for labeling an assignment (Page 45)

Creating objects under any owner

Requirement

- The "2D workspace" area contains objects.
- The objects do not exist in PDMS/E3D.

Procedure

1. To find out which objects do not yet exist in PDMS/E3D, filter by "What's new?". See also chapter Filtering objects (Page 32).
2. Select an object in PDMS/E3D below which one or more missing objects may be created.
3. In the "2D workspace" section, open the shortcut menu of the object that is to be created in PDMS/E3D.
4. Select one of the following commands:
   – "Create (beneath selected)"
     Creates the object below the selected PDMS/E3D object.
   – "Create hierarchy (beneath selected)"
     Creates the object and all underlying objects below the selected PDMS/E3D object.
     If an object cannot be created, the software continues with the next object.

Result

The objects exist in PDMS/E3D and are assigned to the related COMOS objects.

See also

Connection information for labeling an assignment (Page 45)

Displaying potential owner of an object

If you do not know the PDMS/E3D owner below which you may create an object from the "2D workspace" area, you can have all potential owners displayed. You can have several objects that do not yet exist in PDMS/E3D created simultaneously below an owner.

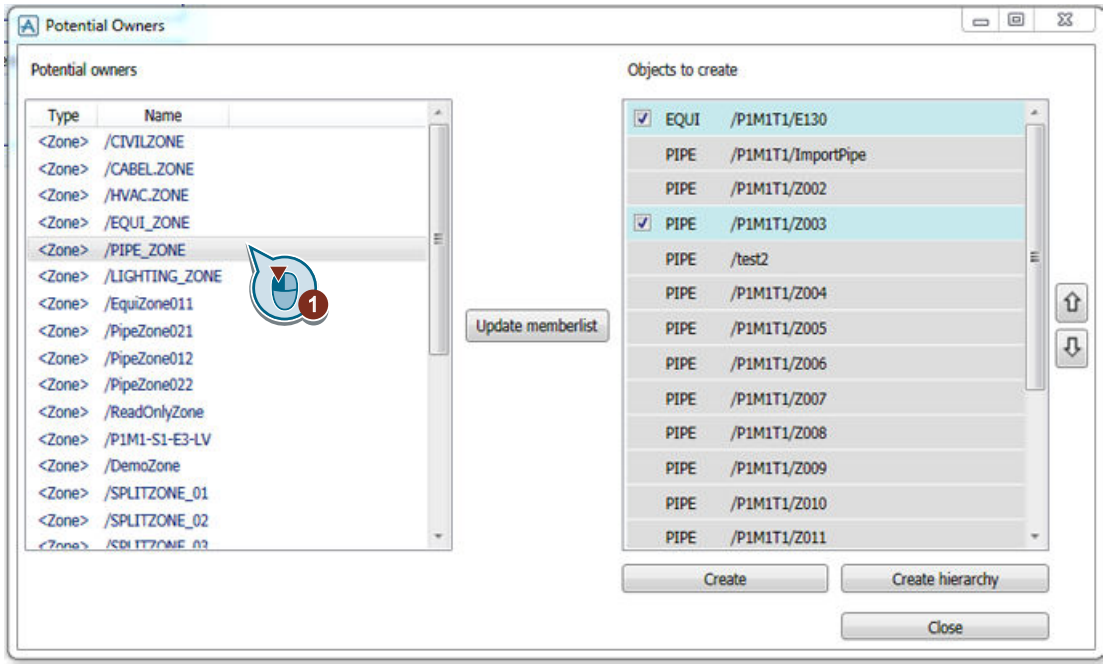
Procedure

1. To find out which objects do not yet exist in PDMS/E3D, filter by "What's new?". See also chapter Filtering objects (Page 32).
2. In the "2D workspace" area, select an object for which you want to display the potential owners.

3. Select the "Show potential owners" command in the context menu.
   The "Potential owners" window opens.

Area	Description
"Potential owners"	Displays all potential owners of the object.
"Objects to create"	Gray: Objects that are already located in PDMS/E3D below the owner that was selected in the "Potential owners" area. Green: Objects that meet the following conditions: • Were requested in the PDMS/E3D Engineering Interface. • Do not exist in PDMS/E3D. • Can be created under the owner which was selected in the "Potential owners" area.

4. Select an owner in the "Potential owners" area.



   The object for which you have selected the "Show potential owners" command as well as all other objects which you could create under the selected owner are highlighted in green and are at the top of the "Objects to create" list. The check box for the objects is selected and means that they will be created in PDMS/E3D when you click "Create" or "Create hierarchy".

5. Optional: Clear the check box of the objects that are not to be created in PDMS/E3D.
6. Optional: To create an object at a specific location under the owner, move the object using the buttons with the up and down arrows.

7. Select one of the following commands:
   - "Create"
     Creates the objects in the specified sequence under the owner.
   - "Create hierarchy"
     Creates the objects and all underlying objects under the searched owner. If an object cannot be created, the software continues with the next object.
8. To update the "Objects to create" area, click "Update memberlist".
   The background color of the created objects changes from green to gray in the "Objects to create" area.
9. Click the "Close" button.

Result

The objects are created in PDMS/E3D in the order in which they are arranged in the "Objects to create" area. They are assigned to the associated COMOS objects.

See also

Connection information for labeling an assignment (Page 45)

Creating pipe components of a virtual pipe section in PDMS

You can select a pipe section in PDMS/E3D and have pipe components created below it that are located below a virtual pipe section in the in the "2D workspace" area of the PDMS/E3D Engineering Interface.

Requirement

- You have requested COMOS objects.
- A virtual pipe section was created in the PDMS/E3D Engineering Interface, which contains pipe components that are to be created in PDMS/E3D.
  See also chapter Requesting COMOS objects (Page 29).

Procedure

1. In PDMS/E3D, select an object of the BRAN type below which the pipe components are to be created.
2. Click the "Create component sequence" entry in the context menu of the virtual pipe section object.

Result

- All pipe components that are located below the virtual pipe section are created below the pipe section in PDMS/E3D. They are generated in the order in which they are arranged on the P&ID.
- The objects are assigned to the corresponding COMOS objects.

See also

Connection information for labeling an assignment (Page 45)

Connection information for labeling an assignment

When you initially assign an object by creating it in COMOS or PDMS/E3D, the objects are assigned to each other. The following properties are entered at the objects:

PDMS/E3D object	COMOS object
• ComosCRefNo • ComosUID • COMOSName If the object was created in COMOS, the name of the object is determined through name mapping in the subclass.	• ComosCheckUID • PDMSRefNo • PDMSName If the object was created in PDMS/E3D, the name of the object is determined through name mapping in the subclass.

The following applies:

- ComosCRefNo = PDMSRefNo
- ComosUID = ComosCheckUID

See also

Filtering objects (Page 32)

Canceling the assignment between COMOS objects and PDMS/E3D objects

Procedure

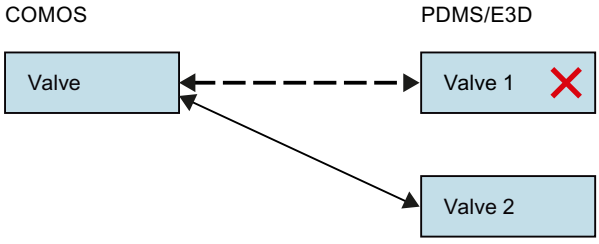
In the COMOS PDMS/E3D Engineering Interface, click the "Unassign" command in the context menu of an assigned object.

Result

- For a correct/unique connection:

  The connection information for the respective COMOS object and PDMS/E3D object is deleted recursively.

- For an incorrect/non-unique connection:



- The connection information for the PDMS/E3D object is deleted recursively. If the COMOS object has a correct connection to another PDMS/E3D object, the correct information is retained.

If you activate the "What's new" filter, the objects are displayed as new objects in which connection information was deleted.

See also

Connection information for labeling an assignment (Page 45)

Assigning COMOS objects and PDMS/E3D objects

You have the following options for assigning objects:

- Assign to the selected object
- Assign by matching names

Requirement

You have requested objects.

Assign to the selected object

1. Select an object in PDMS/E3D.
2. Click the "Assign to selected" entry in the context menu of an object in the "2D workspace" area.

The objects are assigned to each other.

Assign by matching names

1. Select an object in the "2D workspace" area.
2. In the context menu of the object select the "Assign by matching names" command.

If the name of the selected object, which was created through name mapping of the subclass, matches the name of a PDMS/E3D object, the following happens: All COMOS objects below it are recursively assigned to the PDMS/E3D objects below the PDMS/E3D object of the same name.

Result

The assignment causes the following properties to be set at the assigned PDMS/E3D objects:

- ComosCheckUID
- PDMSRefNo
- PDMSName

Updating names in PDMS/E3D

When you create COMOS objects in PDMS/E3D, the names of the objects are formed according to the settings of the corresponding subclass in the "Name mapping" section. You can update the object names to map later changes made to the name mapping.

Procedure

1. Select an object in the "2D workspace" area.
2. Select the "Update > Update names" command in the context menu of the object.
3. To update the names of all objects, click the "Update all names" button in the symbol bar.

Executing COMOS query in PDMS/E3D

You can start a COMOS query in PDMS/E3D and continue to use the query results as needed with the help of a macro you have created yourself.

Requirement

- An administrator has created and configured a subclass for queries. You can find more information on this topic in the "3D Integration Administration" manual, keyword "Creating and configuring query subclasses".
- An administrator has implemented the following PML macro !!SAGPickUpQueryResult(!p is STRING). You can find more information on this topic in the "3D Integration Administration" manual, keyword Implementing PML macro !!SAGPickUpQueryResult(!p is STRING).
- COMOS is open in standard mode or in Design View mode.
- The Engineering Interface is open in COMOS.
- PDMS/E3D is open.

Procedure

1. Call the following PML macro: !!SAGRunCOMOSQuery().
2. Enter the name of a query subclass as argument.

Result

The query results are stored as specified in the macro !!SAGPickUpQueryResult(!p is STRING) and you can edit them further.

3.7 Importing DocLinks

In COMOS, you can create an evaluating report that lists all assigned objects of a 3D model in PDMS Design and COMOS. For this, you export the information for creating DocLinks from PDMS/E3D Draft to COMOS.

Online and offline mode

You can transfer DocLinks to COMOS with two modes:

- If COMOS and PDMS are connected, online mode is enabled.
- If COMOS and PDMS are not connected, offline mode is activated.

3.7.1 Procedure (online and offline)

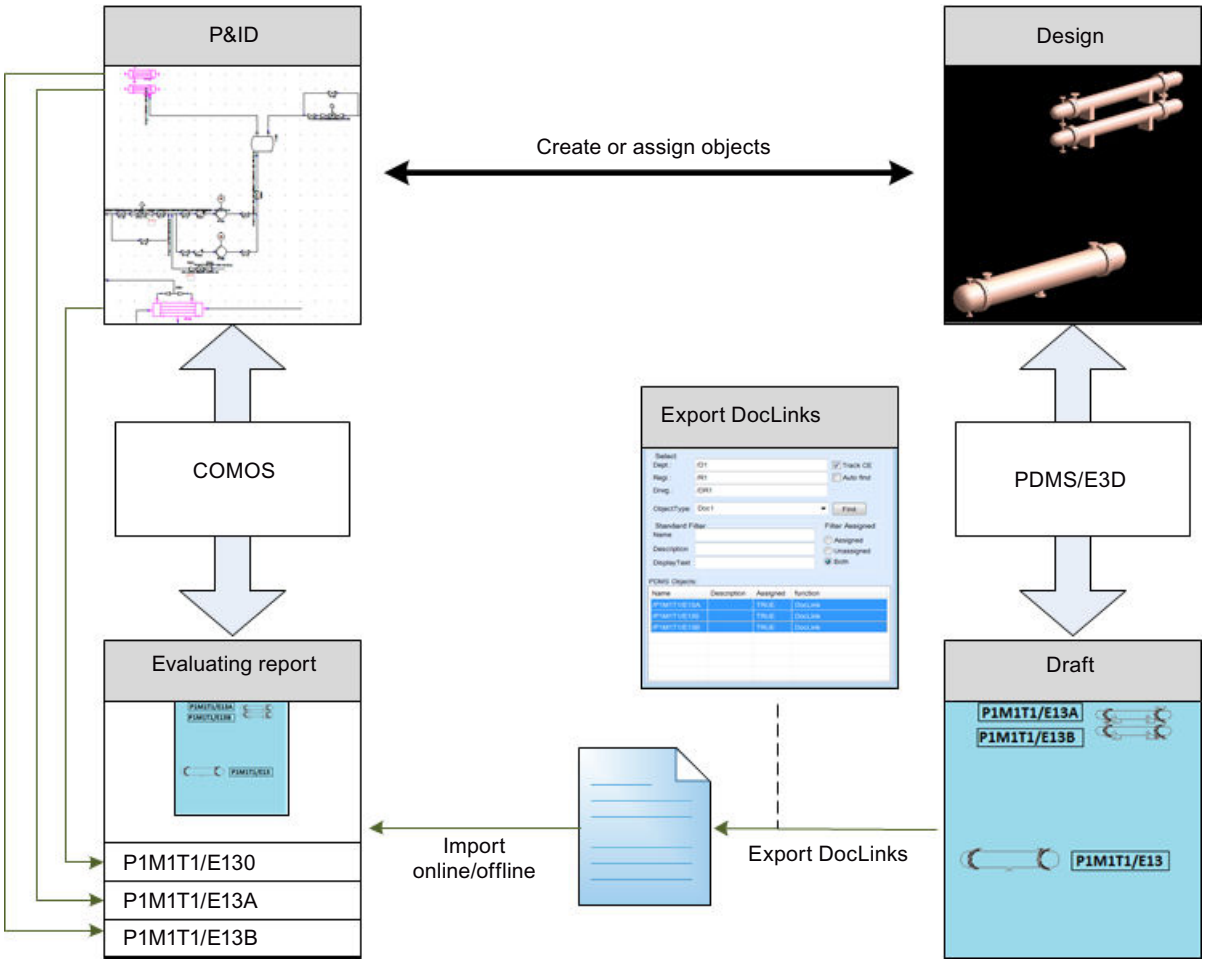
Initial situation: You have associated an object amount from COMOS with your corresponding 3D model.

Requirement

An administrator has configured the import of DocLinks. You can find additional information on this topic in the "3D Integration Administration" manual, keyword "Configuring DocLinks import"".

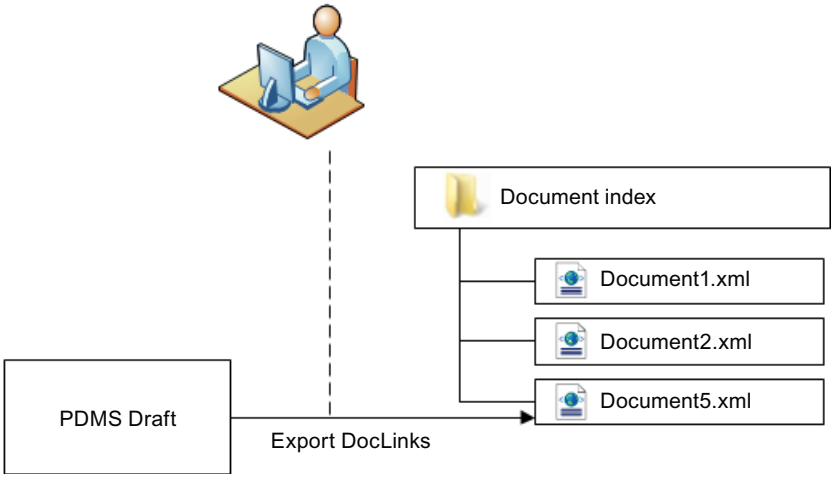
Workflow

Overview of the work sequence:



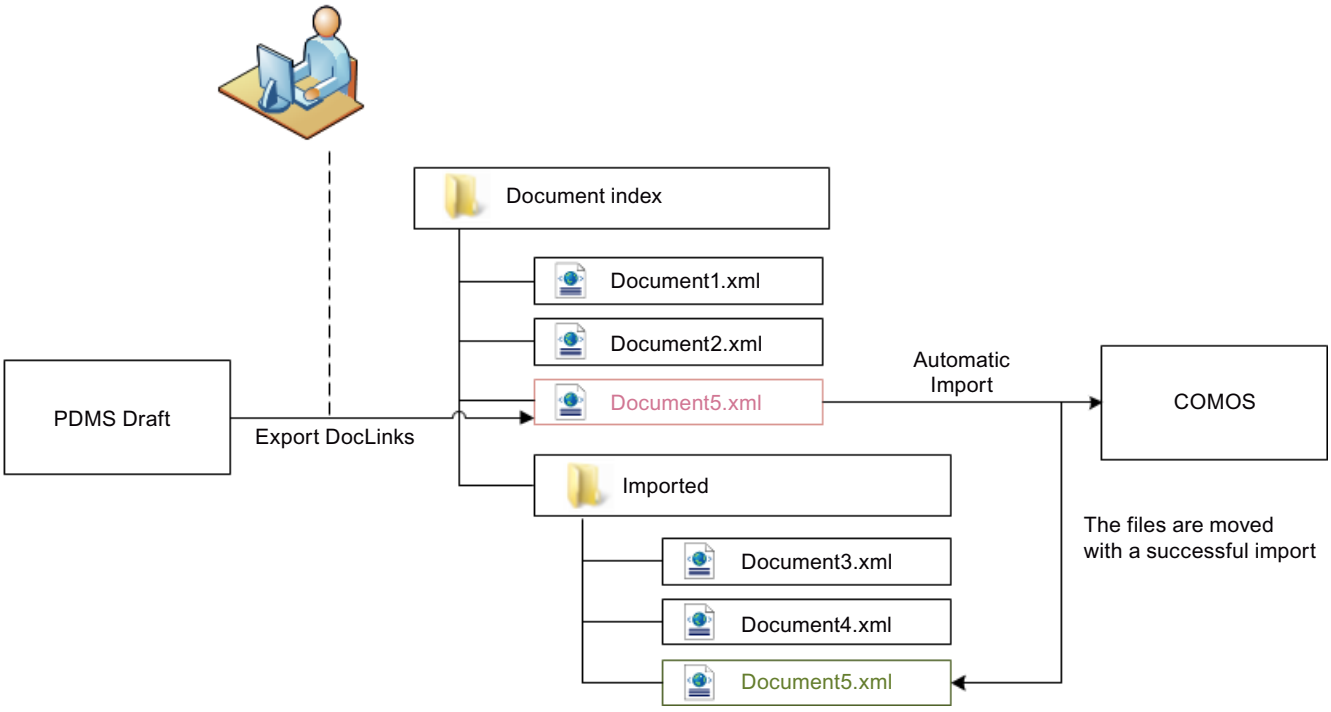
1. Assign the COMOS objects and PDMS/E3D objects to each other in the Design module.
2. Create a drawing in Draft which contains the associated objects.

3. Assign a Label to the objects you have placed (SLAB etc.).
4. Export from labels for the DocLinks to be created in COMOS.
   See also chapter Exporting DocLinks from PDMS/E3D (Page 52).
   During export in offline mode:
   The information is exported to an XML file and stored in the document directory.



During export in online mode:

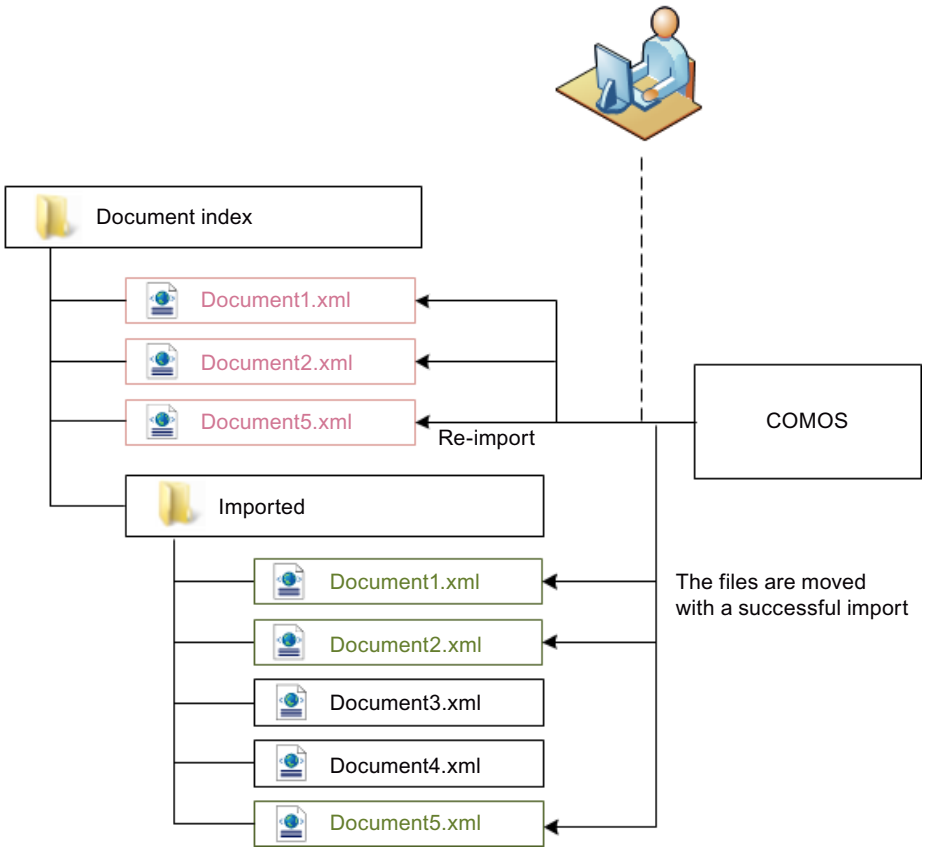
– The XML file is automatically imported into COMOS. A search is made for an evaluating report based on a DocumentID, that an administrator has previously created.
  If the report is found, a DocLink object is created under each object for which a Label exists in the XML file.
  The DocLink objects refer to the evaluating report. The report can be configured so that it displays a list of all DokLink objects as well as a graphic of the Draft layers.
  The XML file is moved to the "Imported" folder.



– If the report is not found, no objects are created. An administrator must create the report. To attempt to read the XML file again and create DocLink objects, proceed as described for import in offline mode.

Import in offline mode:

– Import the DocLinks in COMOS manually. See also chapter Importing DocLinks manually (Page 53).



3.7.2 Exporting DocLinks from PDMS/E3D

Requirement

You have read the next section and the requirements set out there are fulfilled:Procedure (online and offline) (Page 48)

Procedure

1. Open PDMS/E3D Draft.
2. Prepare for the export. See also chapter Exporting DocLinks (Page 101).
3. Open the PDMS/E3D Engineering Interface with the command !!SAGLaunchInterface(). The "Export Doclinks to COMOS" window opens. See also chapter "Export to COMOS" PDMS window (Page 115).
4. Configure the window and export the DocLinks for the desired objects. See also chapter Exporting DocLinks (Page 101).

Result

- If you have exported in online mode and the automatic import to COMOS was successful, DocLinks to the evaluating report are created for all objects for which you have exported labels.
- If the import was not successful, or you have exported in offline mode, import the DocLinks manually. See also chapter Importing DocLinks manually (Page 53).

3.7.3 Importing DocLinks manually

You import DocLinks manually if you exported in offline mode or when the import in online mode has failed.

Import in online mode failed:

- When you run the "Export DocLinks" operation in PDMS Draft, there is no way to always ensure that the specified evaluating report exists in COMOS beforehand. In this case, the XML file remains in the document directory, which is entered in the "PDMS/E3D Engineering Interface" category of the project properties. Otherwise, it is moved to the "Imported" subfolder.
- When you import the DocLinks manually, COMOS tries to import all the remaining documents in the folder XML files again.

Requirement

- You have read the next section and the requirements set out there are fulfilled:Procedure (online and offline) (Page 48)
- An administrator has created an evaluating report. The report has the name specified in the document subclass that you selected when exporting the DocLinks.

Procedure

In COMOS, select the "PDMS Engineering Import > Rerun import" command in the context menu of an object.

Result

The interface attempts to import all XML files from the document directory.

- If the report is found: The DocLinks are created and the file is moved to the "Imported" folder.
- If the report is not found: The XML file is not moved. COMOS does not create DocLink objects.

3.8 Creating material (MTO) from PDMS/E3D

You can create material from PDMS/E3D in the following connection modes:

- Online mode
  COMOS and PDMS/E3D are connected.
- Offline mode
  COMOS and PDMS/E3D are not connected.

3.8.1 Creating material in online mode

You can create material for objects that you have requested in the "3D workspace".

Requirement

- The class "MTO (PDMS)" is configured. You can find additional information on this topic in the "3D Integration Administration" manual, keyword "Configure 'MTO (PDMS)' class".
- The configuration object for material management is created. You can find additional information on this topic in the "COMOS Material Management" manual, keyword "Create configuration object".
  A "Base object for revision folder" is specified in the "General" tab in the properties of the configuration object for material management.
- COMOS and PDMS/E3D are connected.

Procedure

1. Select a pipe in the 3D workspace.
2. Select one of the following commands in the context menu:
   - "Import MTO > Current"
     Select this command for an engineering material that has not been released.
   - "Import MTO > Released"
     Select this command for a released material.

   Optional: Import material for all objects that were requested in the workspace.
   Expand the "Import material for all objects" button.
   - Current
     Imports material that is labeled as current.
   - Released
     Imports material that is labeled as released.

Result

- If the pipe selected in the workspace does not exist in COMOS, it is created.
- A "Current" or "Released" object is created below the pipe.

- The MTO objects are created below the folder.
- If the administrator has configured the class "MTO (PDMS)" accordingly, the pipe lengths from PDMS/E3D were also imported.

3.8.2 Creating material in offline mode

3.8.2.1 Exporting material from PDMS/E3D

Requirement

- You have created and configured the MTOExportSettings.dat file. You can find additional information in the "COMOS Material Management" manual, keyword "MTOExportSettings.dat".
- You have entered the reference numbers of the objects to be generated for recursive material. See also chapter Specifying PDMS/E3D objects for export (Page 57).
- The configuration object for material management is created. You can find additional information on this topic in the "COMOS Material Management" manual, keyword "Create configuration object".
  A "Base object for revision folder" is specified in the "General" tab in the properties of the configuration object for material management. The class 'MTO (PDMS)' is configured.
- The class 'MTO (PDMS)' is configured. You can find additional information on this topic in the "3D Integration Administration" manual, keyword "Configure 'MTO (PDMS)' class".
- You have selected the command "PDMS > Save subclasses" in the context menu for the "Folder for class/subclass definitions (PDMS)" object.

Procedure

Call the following PLM function: !!SAGExportMTO(<parameter>)

Set the following as parameters:

- False: If you are working in IsoDraft
- True: If you are not working in IsoDraft

Result

The material data of the objects specified in the "MTOExportSetting.dat" file is exported. An XML file is created in the Released or Current folder of the exchange directory for each pipe. The folder for the saved file depends on whether "C" or "R" is entered in the first line of the "MTOExportSettings.dat" file.

You can import the XML file in COMOS.

3.8.2.2 Importing material data

Note

Import only possible for assigned objects

You can only create material data for objects that have already been created and assigned in COMOS. When you import material for objects that do not have a valid assignment, no material is created.

Requirement

- You have created and configured the MTOExportSettings.dat file. You can find additional information in the "COMOS Material Management" manual, keyword "MTOExportSettings.dat".
  You have entered the reference numbers of the objects to be generated for recursive material.
- The configuration object for material management is created. You can find additional information on this topic in the "COMOS Material Management" manual, keyword "Create configuration object".
  A "Base object for revision folder" is specified in the "General" tab in the properties of the configuration object for material management.
- The class "MTO (PDMS)" is configured.
  You can find additional information on this topic in the "3D Integration Administration" manual, keyword "Configure 'MTO (PDMS)' class".
- You have selected the command "PDMS > Save subclasses" in the context menu for the "Folder for class/subclass definitions (PDMS)" object.
- You have exported material from PDMS/E3D. See also chapter Exporting material from PDMS/E3D (Page 55).

Procedure

1. Select any object in the COMOS Navigator.
2. In the context menu, select the command "MTO> From PDMS/E3D > MTO Transfer".
   A window opens in which you can select the folder with the XML files. By default, the XML files are saved in the exchange directory.
3. Open the exchange directory.
   You determine the exchange directory in the project properties in the "Exchange directory" field of the "PDMS/E3D Engineering" category.
4. Confirm your entries.

Result

- All XML files in the selected folder are imported.
- Material objects are created under all the assigned objects for which you have imported material. If the administrator has configured the class "MTO (PDMS)" accordingly, the pipe lengths from PDMS/E3D were also imported for pipes.
- The values of the attributes specified in the name directory are transferred to the properties of the MTO objects.

3.8.2.3 Specifying PDMS/E3D objects for export

Procedure

Expand the content of the MTOExportSettings.dat configuration file by adding the names of the pipes to be exported:

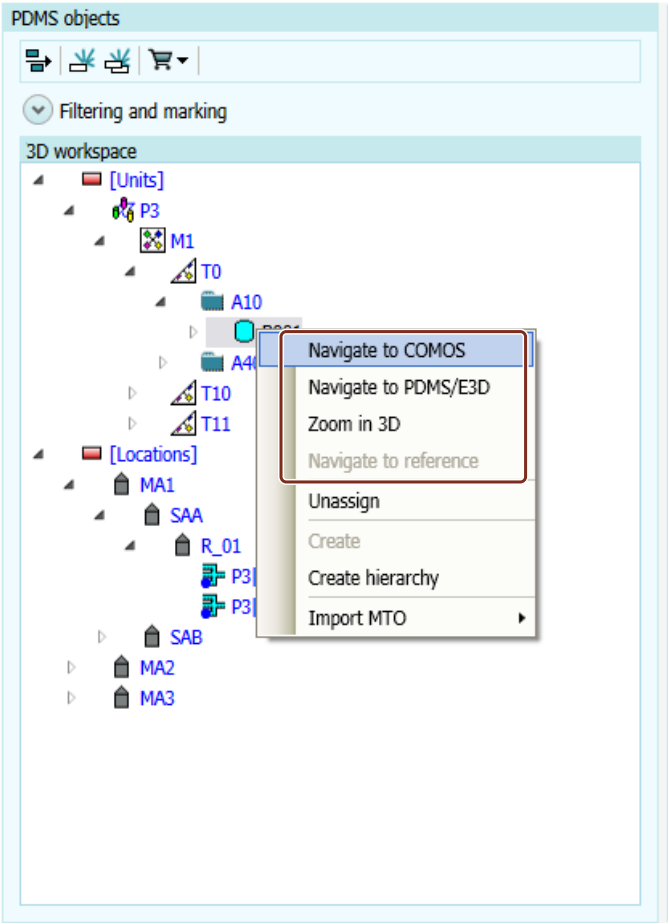
```
// export status of pipes / branches: set R for "released", set C for
"current"
// path to the material files
// path to the current files
// path to the released files
// path to the temp after iso files
// path to the subclass file and the name of it
// The following code will be created automatically - do not modify
it!
// [!START!] The following objects will dealed:
=8261/19717
=8261/19730
=8261/19753
```

The pipes are specified by your `RefNo`. Here you can also specify the `RefNo` of a `ZONE` or `SITE`.

3.9 Navigating

3.9.1 Navigating from the 3D workspace

You have several options for navigating from the "3D workspace".



Navigating to the COMOS Navigator

Requirement:

- You have created the object in COMOS.
- The COMOS object and PDMS/E3D object are assigned to each other.

Click "Navigate to COMOS" in the shortcut menu of the object.

Navigating to the PDMS/E3D Navigator

Click "Navigate to PDMS/E3D" in the shortcut menu of the object.

Zooming in on an object in PDMS/E3D in 3D model

Requirement: The object exists in the 3D model in PDMS/E3D.

Click "Zoom in 3D" in the shortcut menu of the object.

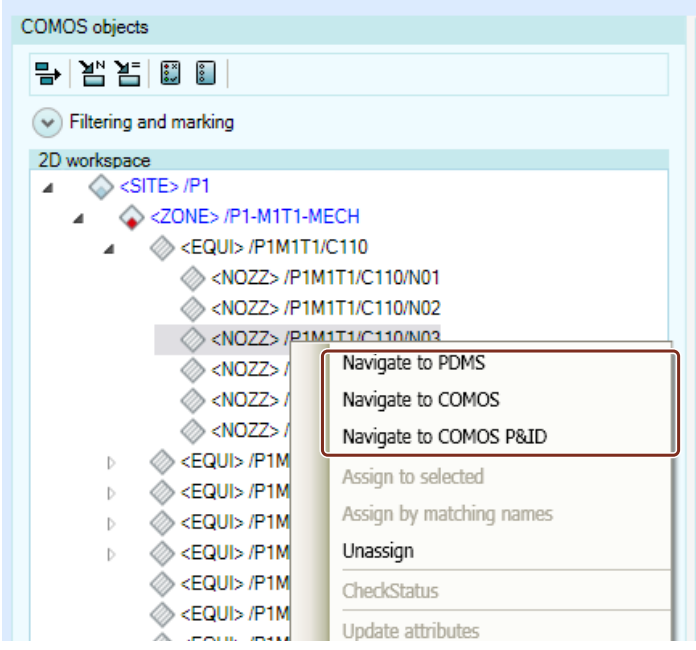
Navigating to reference

This command is only activated when the object has a reference to another object.

Click "Navigate to reference" in the shortcut menu of the object.

3.9.2 Navigating from the 2D workspace

You have several options for navigating from the "2D workspace".



Navigating to the PDMS/E3D Navigator

Requirement:

- You have created the object in PDMS/E3D.
- The COMOS object and PDMS/E3D object are assigned to each other.

Click "Navigate to PDMS/E3D" in the shortcut menu of the object.

Navigating to the COMOS Navigator

Click "Navigate to COMOS" in the shortcut menu of the object.

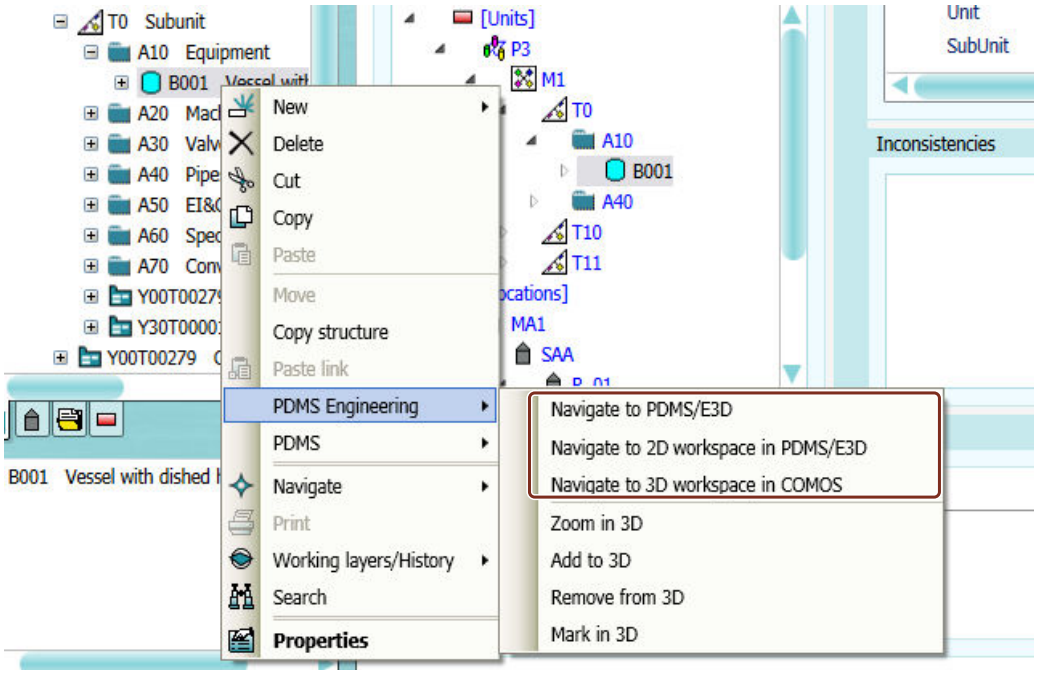
Navigating to the object in COMOS P&ID

Requirement: The object is placed on a P&ID.

Click "Navigate to COMOS P&ID" in the shortcut menu of the object.

3.9.3 Navigating from the COMOS Navigator

You have several options for navigating from the COMOS Navigator.



Note

Navigating from pipe segments

From a pipe segment you can navigate only to its owner in PDMS/E3D or in the "2D Workspace" area.

Requirement

The COMOS object has been assigned to a PDMS/E3D object.

Navigating to the PDMS/E3D Navigator

1. Select an object in the Navigator.
2. Select "PDMS Engineering > Navigate to PDMS/E3D" in the shortcut menu.

The assigned object is selected in PDMS/E3D.

Navigating to the 2D workspace in PDMS/E3D

1. Select an object in the Navigator.
2. Select "PDMS Engineering > Navigate to 2D workspace in PDMS/E3D" in the shortcut menu.

The assigned object is selected in PDMS/E3D.

Navigating to the 3D workspace in COMOS

1. Select an object in the Navigator.
2. Select "PDMS Engineering > Navigate to 3D workspace in COMOS" in the shortcut menu.

3.9.4 Navigating from the PDMS/E3D menu

If COMOS and PDMS/E3D are connected via the Interface, you can find additional navigation options in the "COMOS" menu.

Requirement

You have selected an assigned object in the PDMS/E3D Design Explorer.

Procedure

- "COMOS > Navigate to COMOS"
  Navigate to the COMOS Navigator
- "COMOS > Navigate to COMOS P&ID"
  If the object is placed in an P&ID, the P&ID opens and the object is selected.
- "COMOS > Navigate to 2D workspace"
  Navigate to the section "2D workspace" in PDMS/E3D.
- "COMOS > Navigate to 3D workspace"
  Navigate to the section "3D workspace" in COMOS.

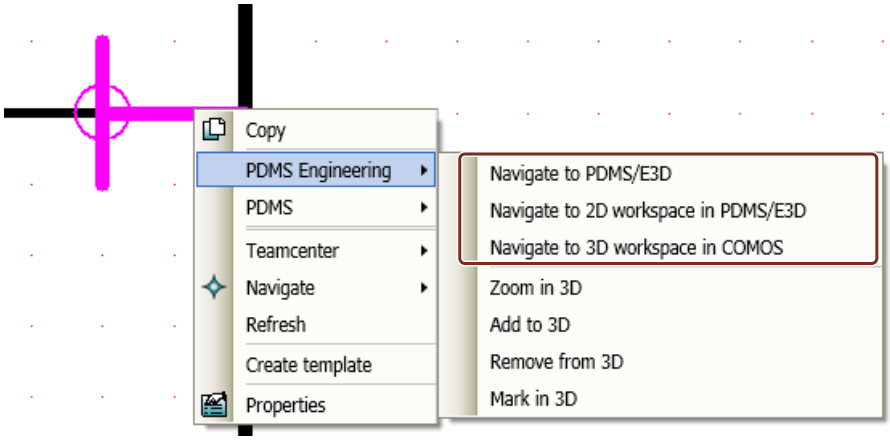
3.9.5 Navigating from P&IDs

You have several options for navigating from a P&ID.

Note

Navigating from pipe segments

From a pipe segment you can navigate only to its owner in PDMS/E3D or in the "2D workspace" area.



Navigating to the PDMS/E3D Navigator

Requirement: The COMOS object and PDMS/E3D object are assigned to each other.

In the shortcut menu of the object on the P&ID, click the "Navigate to PDMS/E3D" command.

Navigating to the 2D workspace in PDMS/E3D

Requirement: The COMOS object and PDMS/E3D object are assigned to each other.

In the shortcut menu of the object on the P&ID, click the "Navigate to 2D workspace in PDMS/E3D" command.

Navigating to the 3D workspace in COMOS

In the shortcut menu of the object on the P&ID, click the "Navigate to 3D workspace in COMOS" command.

3.9.6 Navigating between the reference and original object

Requirement

You have created objects in COMOS that have a reference to other objects.

Navigate to original

1. Select a referenced object in the "Units" or "Locations" node in the "3D workspace" area. Referenced objects are identified by a blue dot. Example:

2. Select the "Navigate to original" command in the context menu.

Navigate to reference

1. Select an object in the "Units" or "Locations" node in the "3D workspace" area that has a reference to another object.
2. Select the "Navigate to reference" command in the context menu.

3.9.7 Using 3D operations

Procedure

1. Select an object in the Navigator or on a P&ID.
2. Select the "PDMS Engineering > 3D View > <Operation>" entry in the context menu.
   The following operations are offered depending on the object:
   – "Zoom in 3D"
     See chapter "Zoom" operation (Page 94)
   – "Add to 3D"
     See chapter "Add" operation (Page 92)
   – "Remove from 3D"
     See chapter "Remove" operation (Page 93)
   – "Mark in 3D"
     See chapter "Mark" operation (Page 93)

COMOS PDMS interface 4

4.1 General information on COMOS PDMS Interface

Aim

COMOS PDMS Interface is an interface that enables you to integrate data generated in COMOS into the 3D planning of AVEVA PDMS.

Interface operations:

- You use this interface to import and export your construction objects and synchronize P&ID and 3D data.
- On the COMOS side, you use the COMOS Design Viewer to work with the interface. The COMOS Design Viewer is a version of COMOS limited in functionality.
- On the PDMS side you use the commands from the "COMOS" menu to operate the interface, working in the "Export to COMOS" and "Query COMOS" windows.
- The data exchange is configured in COMOS.

The implementation of the interface is based on algorithms that you can use to adapt the default settings of the supplied database to the specific requirements of your company.

The COMOS PDMS interface is XML-based and is suitable for both local installation and installation in a Citrix environment.

Functional scope of interface operations

The interface covers the following functionalities:

- Export of construction objects from COMOS to PDMS
- Import of construction objects from PDMS to COMOS
- Definition of the owner restriction
- Navigating between the COMOS objects and the corresponding PDMS objects.
- Use of various 3D view operations: Zoom, Add, Remove, Mark
- Implementation of status checks for the objects in both applications.
- If a status check shows that new objects have been created or the attributes of objects that have already been connected have changed: Synchronization of the data by importing missing objects or synchronizing existing objects.
- Assignment of COMOS objects and PDMS objects to one another and canceling assignments.

4.2 Requirements

The following requirements have to be met in order to use COMOS PDMS Interface.

If one of the requirements is not met, contact your administrator.

Installed software

- The software requirements are met. You can find additional information on this topic in the "3D Integration Administration" manual, keyword "Software requirements".
- The "COMOS PDMS Interface" is installed and activated. You can find additional information on this topic in the "3D Integration Administration" manual, keyword "Activating COMOS PDMS Interface for COMOS".

COMOS database

- All base data and standard tables required for the interfaces have been imported to the database.
- The configuration for the interface operations area has been completed.
- The names of the PDMS pipe specs are specified in the standard tables.
- The following requirements are met:
  - The project properties have been fully configured.
  - The interface has been activated for the project.
    If the interface has not been activated, activate it. See also chapter Opening the COMOS PDMS Interface (Page 66).
  - By creating the construction objects and P&IDs you have starting the planning work.

In PDMS

The necessary UDAs are available.

4.3 Opening the COMOS PDMS Interface

The interface is available with the following PDMS modules:

- Design
- Draft
- Paragon

Requirement

See chapter Requirements (Page 66).

Procedure

1. Start one of the above named modules.
2. Open the command line.
3. Enter the following call in the command line: "!!ITSetupComos()"
   - Design and Draft modules:
     The "COMOS" menu is available in the PDMS menu bar.
   - Paragon module: The interface is started.
4. Start COMOS PDMS Interface in the Design and Draft modules using the command: "COMOS > Interface > Start interface.

Note

Perform these steps again at each start of PDMS or module change.

Deactivating COMOS PDMS Interface

In PDMS, in the Design and Draft modules:

1. Start the corresponding module.
2. Select the "COMOS > Interface > Stop interface" command from the menu.

4.4 Operating mode

Online mode and offline mode

The following operating modes are available for COMOS PDMS Interface:

- Online mode
  - PDMS and the COMOS Design Viewer or COMOS are open.
  - The interface is activated.

  Available for the plug-ins "PDMS/E3D Engineering Interface" and "PDMS/E3D Pipe Specs Import".
  The other program in each case is automatically informed that certain operations can be executed.
- Offline mode
  - Only one of the applications is open.
  - The interface operations area is not available.

  You must start operations manually.

In both cases, COMOS and PDMS communicate via XML files that are stored in the communication path and in the exchange path and contain the necessary information.

Note

This documentation describes online mode.

4.5 Using interface operations

Availability of interface operations

Some interface operations are called within COMOS, others in PDMS. Some interface operations are available in both COMOS and PDMS.

The following section contains an overview of which interface operations exist and in which application they are available.

COMOS interface operations

- "Export" and "Export to CE":
  Exports COMOS objects to PDMS.
- "Refresh" and "Custom Refresh":
  Refreshes the attributes of objects which are interconnected in COMOS and PDMS.
- "Assign":
  Connects objects that are available in both systems.
- "Add":
  Adds objects to the PDMS drawlist.
- "Remove":
  Deletes objects from the PDMS drawlist.
- "Zoom":
  Performs a zoom operation for objects in PDMS.
- "Mark":
  Marks objects in PDMS.
- "Rerun documents":
  If the "Import DocLinks" operation could not be executed successfully for documents in PDMS, this operation repeats the import.

Interface operations that are called in PDMS

- "Export":
  Exports PDMS objects to COMOS.
- "Import DocLinks":
  Places COMOS objects from a PDMS draft drawing onto a COMOS document.
- "Query COMOS data":
  Runs a query in COMOS and displays the result set in PDMS.

Interface operations that are called both in COMOS and in PDMS

- "Unassign":
  Releases existing connections between COMOS and PDMS objects.
- "Check status":
  Checks the status of connected objects.
- "Select"/"Navigate":
  Sets the current element in PDMS and navigates to the COMOS object connected to it.

4.5.1 Calling interface operations

There are various procedures for calling an interface operation, depending on the respective application.

Calling COMOS interface operations

1. Select an interface object.
2. Select the required operation from the "PDMS > ..." context menu.
   You can call this context menu from various places in COMOS:
   - In the Navigator
   - On a P&ID
   - In a query

   You can call the context menu for a single object (single selection) or for multiple objects (multiple selection).

Calling PDMS interface operations

In PDMS you can call interface operations from the following places:

- For the current element:
  Via the PDMS menu bar, "COMOS > CE > ..." menu.
- For one to any number of objects specified in the "Export to COMOS" window:
  Via the context menu in the result list in the "Export to COMOS" window.
- The "Query COMOS data" operation:
  Via the PDMS menu bar, "COMOS > Interface > Query COMOS data" menu, "Query COMOS" window.

See also

Working in the "Export to COMOS" window (Page 75)

Interface operations in COMOS (Page 80)

PDMS-side interface operations (Page 95)

To create a selection set, call an interface operation. The interface operation is applied to the objects contained in the selection set.

Which objects belong to the selection set depends on whether you called the interface operation from COMOS or from PDMS.

4.5.2 Selection set for COMOS interface operations

Overview

If you call the interface operation from COMOS, the creation of the selection set depends on the point from which the context menu was called.

- In the Navigator
- On a P&ID
- In a query
- For a single object, i.e. using single selection
- For multiple objects, i.e. using multiple selection

Call In the Navigator for a single object

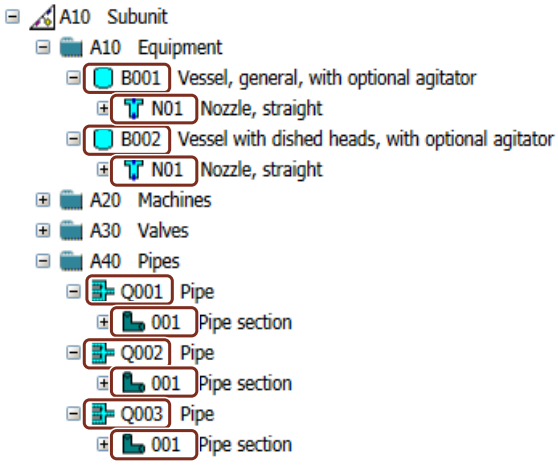
If you select a single object in the Navigator and select an interface operation in the context menu of the object, the selection set is created as follows:

- All objects located underneath the selected engineering object are processed recursively.
- All interface objects found during this run will be added to the selection set.
  Definition of interface objects: You can find additional information on this topic in the "3D Integration Administration" manual, keyword "Interface object".

- If the in the Navigator selected object is also an interface object, it is also added to the selection set.
- If the object selected in the Navigator is a P&ID, all objects placed on the P&ID are inserted in the selection set.

Example:

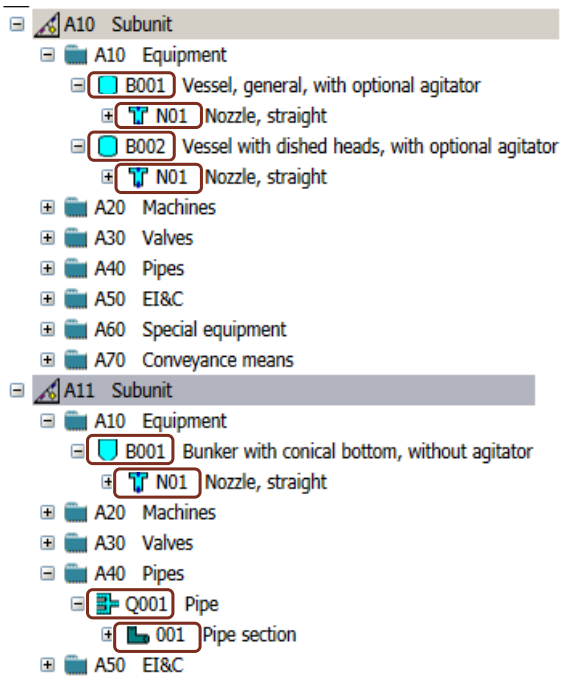
If you select the subunit and select an interface operation, the selection set shown on the right will be created. Requirement: Vessels, nozzles, pipes and pipe sections are interface objects.



Call in the Navigator for multiple objects

If you select multiple objects in the Navigator and then select an interface operation in the context menu, the selection set is generated in the same way as when a single object is called:

Example:

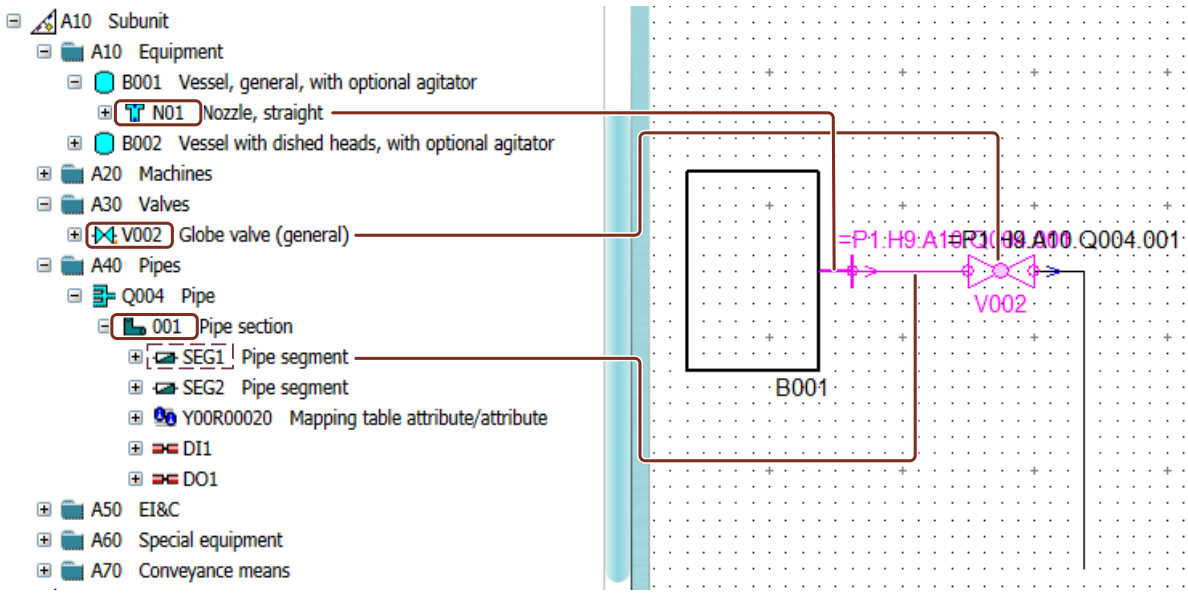


Call on a P&ID for one or multiple objects

If you select one or more objects on a P&ID and then select an interface operation in the context menu, the selection set is created as follows:

- All interface objects from the selected objects are added to the selection set.
- If the selected object is a pipe segment and its pipe section is an interface object, the pipe section is added to the selection set.
  If the pipe section belongs to the "Undef" class and the owner is an interface object, the owner is added to the selection set.

Example:



Call for a complete P&ID

To call an interface operation for a complete P&ID, you have the following options:

- Select a P&ID in the Navigator and select an interface operation in the context menu.
- Select an interface operation in the context menu on an open P&ID without an object selected on the P&ID.

The selection set is generated as follows:

All interface objects placed on the P&ID are added to the selection set.

Call in a query

If you use a query to perform an interface operation, select any number of objects in the query. Once you have selected the required objects, select an interface operation in the context menu.

The same selection set is generated as after selection of the objects in question in the Navigator.

Location objects are handled separately. Depending on the operation, the objects are added to the selection set on the location or unit side. If you call an interface operation on the location tree, the location objects are always added to the selection set.

4.5.3 Selection set for PDMS interface operations

Which objects are part of the selection set depends on where you called the interface operation:

- Via the "COMOS > CE" menu:
  If the current element is an interface object, only the current element is part of the selection set.
- Via the context menu of the result list in the "Export to COMOS" window.
- Via the "COMOS > Interface > Query COMOS data" menu.

Selection set via "Export to COMOS"

The "Export to COMOS" window appears if you have selected the "COMOS > Interface > Export to COMOS" menu.

In the window you define which PDMS objects belong to the selection set.

Selection set via "Query COMOS data"

The "Query COMOS" window appears if you have selected the "COMOS > Interface > Query COMOS data" menu.

In the window you define the COMOS objects for which a query is to be started from PDMS.

See also

PDMS: "Export to COMOS" window (Page 75)

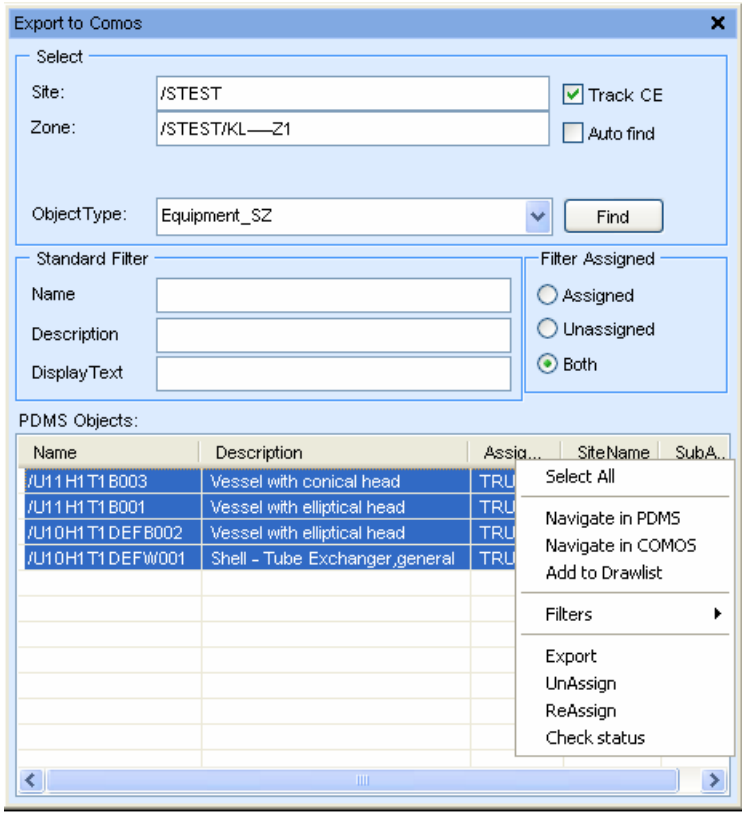
PDMS: "COMOS query" window (Page 78)

4.5.4 PDMS: "Export to COMOS" window

4.5.4.1 Working in the "Export to COMOS" window

Purpose

In the "Export to COMOS" window, you set various filters and options to define a set of PDMS interface objects. In a further step you then execute an interface operation.



Note

Only objects that meet one of the following conditions are considered here:

- Objects to which a subclass has been assigned
  This means that the ":ComosSClass" UDA must contain the name of a valid subclass as a value.
  Reason: The objects that are added to the selection set are filtered on the basis of their ":ComosSClass" UDA.
- Objects whose PDMS element type is the same as one of the PDMS element types that were entered in COMOS at the subclass selected in the "Object Type" field.
  During the operation, the interface automatically sets the ":ComosSClass" UDA at these objects.

Procedure

To work in the "Export to CE" window, proceed as follows:

1. Select the following menu in PDMS: "COMOS > Interface > Export to CE"
   Result: The "Export to CE" window opens.
2. Configure the "Select" area.
   See also chapter Configuring the "Export to COMOS" window (Page 77).
3. Optionally: Configure the following filters:
   – The filters of the "Standard Filter" area
   – The filters of the "Filter Assigned" area
     See also chapter PDMS: "COMOS query" window (Page 78).
4. Click the "Find" button in the "Select" area.
   Result:
   – Your entries and filters are evaluated.
   – A result list is displayed in the "PDMS Objects" area.
5. Optionally:
   – Right-click anywhere in the result list in the "PDMS Objects" area.
   – Select the "Filters > …" command from the context menu and filter the result list on the basis of the values offered in the submenus.
     Information regarding the configuration of these filters: see the link below.
6. Mark one or more objects in the result list and call the context menu.
7. Select one of the following commands from the context menu:
   – "Export"
   – "Unassign"
   – "Reassign"
   – "Check status"

Other functions of the "Export to COMOS" window

If object are already displayed in the result list, the context menu of the list will also offer the following functions:

- "Select All":
  Selects all objects in the result list.
- "Navigate in PDMS":
  Activated: When you select an object in the result list, it is simultaneously selected in the Design Navigator in PDMS.

- "Navigate in COMOS":
  Activated: When you select an object in the result list, the "Select" interface operation is called simultaneously.
  Result: If the object is connected with a COMOS object, the connected COMOS object is also selected in COMOS.
- "Add to Drawlist":
  Adds the currently selected objects to the PDMS Drawlist.

This function makes it easier for you to work with the window or with the interface.

See also

"Export to COMOS" PDMS window (Page 115)

4.5.4.2 Configuring the "Export to COMOS" window

The "Export to COMOS" window consists of the following control groups, in which you can make settings. See also chapter "Export to COMOS" PDMS window (Page 115).

- "Select"
- "Standard Filter"
- "Filter Assigned"
- "PDMS Objects"

Procedure

1. In the "Select" control group, specify the basic settings for filtering the result list.
2. In the "Standard Filter" control group, filter the result list on the basis of specific texts.
3. In the "Filter Assigned" control group, filter the result list based on the connection information.
4. In the "PDMS Objects" control group, filter the attribute columns of the result list on the basis of specific values. See also chapter Filtering the attribute columns of the result list (Page 77).

4.5.4.3 Filtering the attribute columns of the result list

Procedure

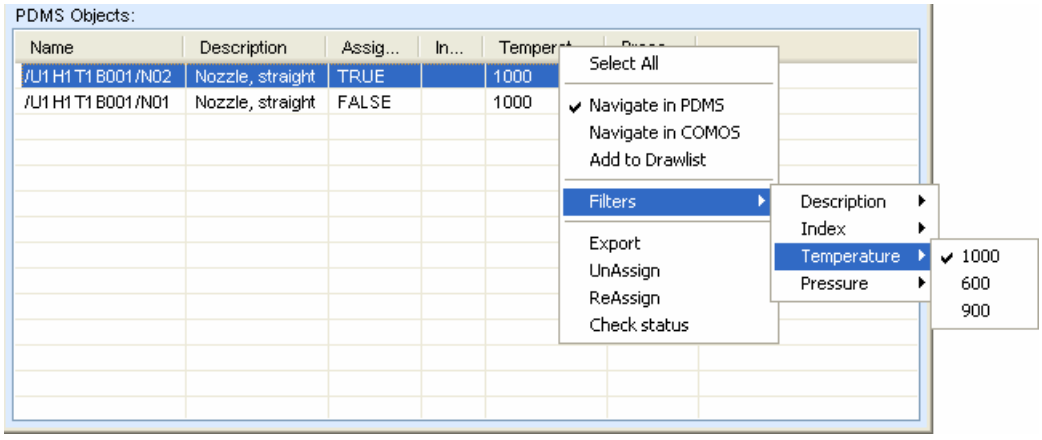
1. Right-click on any location in the result list
2. Select the "Filter > ..." command from the context menu.
   All attributes your administrator has set as filters are offered in the submenu.
3. Select an attribute.
   The submenu displays the values which the objects currently contained in the list accept for the selected attribute.

4. Select one or more values.
5. Repeat these steps for any number of attributes.

Result

Only the objects that have the attribute values you selected are visible in the result list.

Example



4.5.5 PDMS: "COMOS query" window

4.5.5.1 Configuring the "Query COMOS" window

Procedure

1. Select the "COMOS > Interface > Query COMOS data" command from the PDMS menu bar.
   Result: The "Query COMOS" window opens.
2. Configure the "Query" tab.
   See also chapter "Query COMOS" PDMS window (Page 117).
3. Configure the "Filters" tab.
   See also chapter "Query COMOS" PDMS window (Page 117).
   See also chapter Defining filters (Page 79).
4. Configure the "Execute" tab.
   See also chapter "Query COMOS" PDMS window (Page 117).

4.5.5.2 Defining filters

Procedure

1. Open the "Filters" tab.
2. Select a line in the "Columns" area.
3. Select the "Add filter for this column" command from the context menu.
   Result: The "Filter" area is activated.
4. Select an operator from the "Operator" list in the "Filter" area and enter a filter criterion in the "Criteria" field.
   Information regarding operators: See below.

Result

- The filter is added to the "Filters" area:
  - "Column" column: The name of the column for which you defined a filter
  - "Operator" column: The operator you defined for this column
  - "Criteria" column: The criterion you defined for this column
- The operator and the criterion are applied to every line of the query. The line is not added to the result file if the value of the column entered in "Column" does meet the filter criteria.

Removing filters

1. Open the "Filters" tab.
2. Select a line in the "Columns" area.
3. Select the "Remove filter for this column" command from the context menu.

Result

All filters defined for this column are removed.

4.5.5.3 Executing the COMOS query

Procedure

See chapter "Query COMOS data" operation (Page 100).

Result

Once the query has been executed in COMOS, the result of the operation is displayed on the "Results" tab in the "Query COMOS" window. See also chapter "Query COMOS" PDMS window (Page 117).

4.6 Interface operations in COMOS

4.6.1 "Export" operation

Purpose of the "Export" operation

The "Export" operation creates objects for the objects in the selection set in PDMS and connects them to the COMOS objects.

Requirements in PDMS

- In order to create an object in PDMS, the following conditions must be met:
  - The corresponding owner restriction rules must be met.
  - You have write rights in PDMS which enable you to create an object underneath a specified object.
  - The object underneath which the object to be created is supposed to be generated must exist.
  - The object to be exported must not yet exist on the PDMS side.
- In the case of single selection, you can place the object in 3D space by determining a position point in PDMS (ID @) with the help of the cursor.
  For this, the following conditions must be met:
  - The object is not a pipe component.
  - The object has a position (attribute position).

  Note

  The "Export" and "Export to CE" interface operations in COMOS do not create any Sites in PDMS. However, zones are created if necessary when Export is called.

Call

To call the "Export" interface operation, select the "PDMS > Export" or "PDMS > Export to CE" command in the context menu of an object in the Navigator.

Defining objects in the selection set

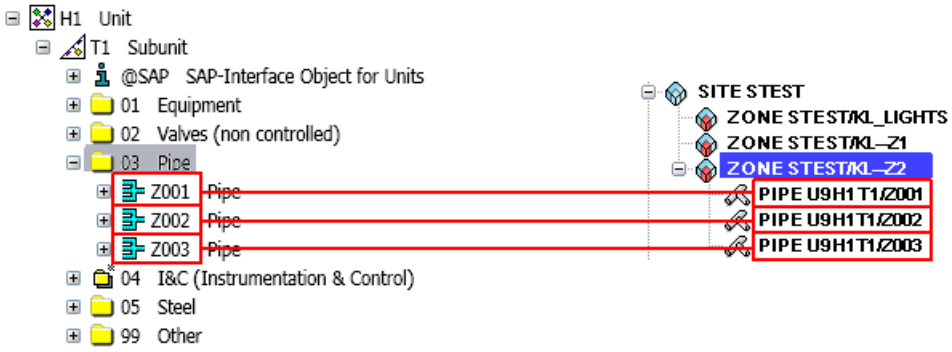
You can use the submenu of the context menu to select whether you only want to export objects from the "Tagged item" class or from the "Cable" class:

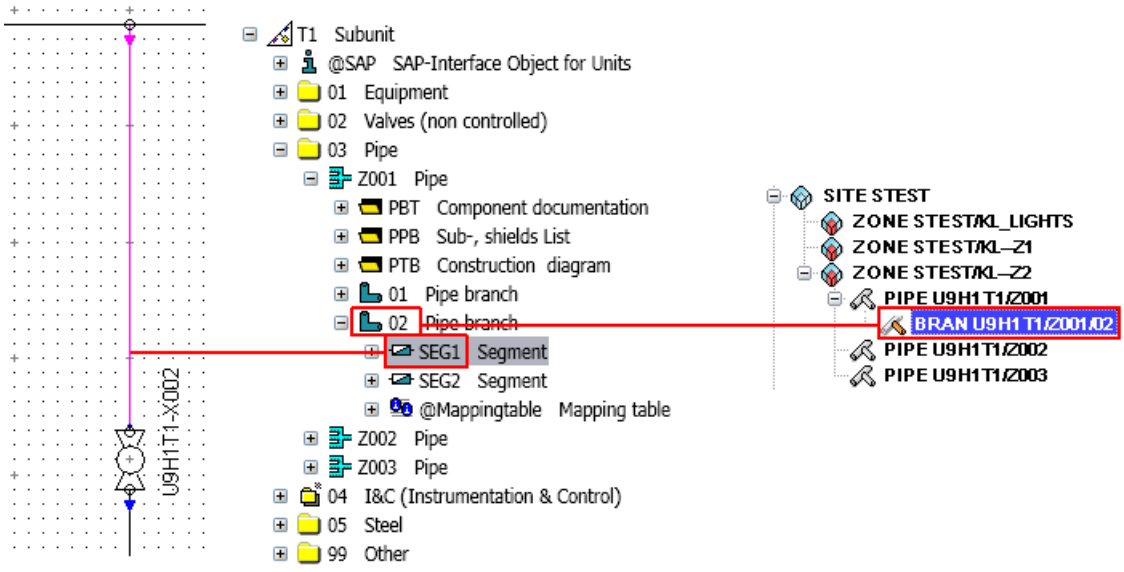
- Submenu "... Export > Equipment":
  When you select this submenu, all objects in the selection set from the "Tagged item" class are processed.
- Submenu "... "... Export > Cable":
  When you select this submenu, all objects in the selection set from the "Cable" class are processed.

If, after an initial verification, the selection set can only consist of objects from the "Tagged item" or "Pipe" class, only the "... > Tagged item" or "... > Cable" command is offered in the context menu.

Furthermore, you can use the "Options > Export recursive" command in the context menu to define whether or not the interface objects located underneath interface objects in COMOS are also added to the selection set.

This functionality allows you, for example, to first export the pipes and then to transfer the pipe sections to PDMS via the P&ID at a later point in time.





Note

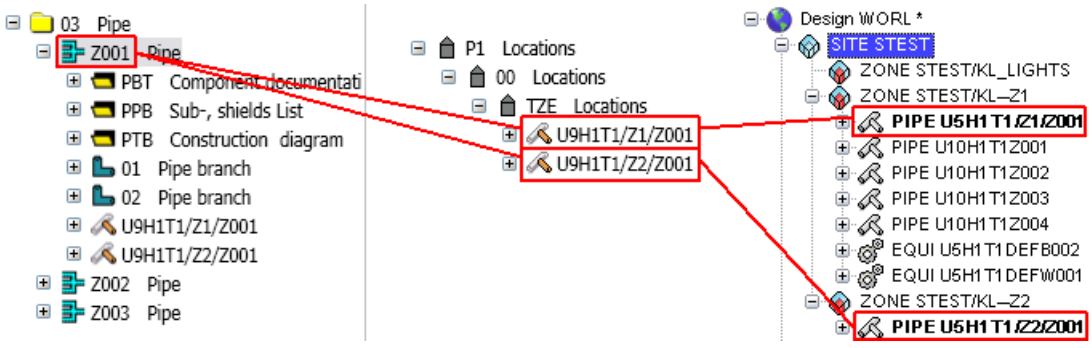
If the class of the pipe section is "Undef", only pipes from the P&ID can be exported to PDMS.

Location objects

Location objects are added to the selection set if the operation was started in the location view. If the "Export" operation is called for an object which is configured so that location objects are created, the following applies:

Objects are only created in the COMOS location if the corresponding PDMS object could be created. The object in the location view is connected with the PDMS object.

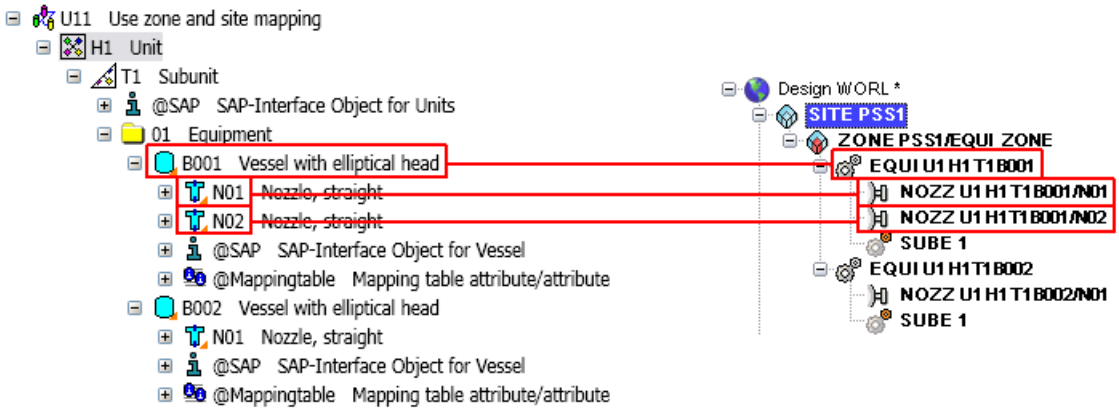
For example, if pipe Z001 creates location objects, the following situation will arise if the pipe has been created twice under different zones in PDMS:



Retaining the object structure

The "Export" operation preserves the structure, meaning that direct hierarchical relations in COMOS are mapped in PDMS.

For example, if vessels and nozzles are interface objects, when the vessel is exported, the nozzles of the vessel are exported as well. In PDMS, the nozzles are then also created underneath the PDMS object created for the vessel.



Insertion location in PDMS

When you run the "PDMS > Export to CE" command, the objects from the selection set are created below the current element in PDMS.

When you run the "PDMS > Export" command, the PDMS object below which the objects are to be created is determined as follows:

- If the object is to be created in PDMS underneath a zone, the name of the PDMS object is generated through a mapping defined in the configuration.
- If the owner is not a zone, the potential owner in PDMS can be determined by checking if the owner in COMOS is connected to a PDMS object.
  If this is the case, the object is created underneath this PDMS object. This rule is not applied when the owner of the object is contained in the selection set. In that case the hierarchical relation in COMOS takes precedence.

Result in PDMS

- The corresponding PDMS objects are created if this is possible.
- The objects that were created in PDMS are connected to the corresponding COMOS objects.
- Data exchange is performed at attribute level for the objects that were just connected.
- If an object already exists in PDMS, it is not connected and no data exchange is carried out at attribute level either.
- Afterwards, the attributes of the connected objects which are relevant for a status check are compared with one another as specified in the configuration. See also chapter "Check status" operation (Page 89).
- A status is calculated for all objects in the selection set.

4.6.2 "Refresh" operation

Purpose of the "Refresh" operation

The "Refresh" operation updates already connected objects at the attribute level.

Call

To call the "Refresh" interface operation, select the "PDMS > Refresh" command from the context menu of an object in the Navigator.

Note

Refresh attributes configuration

Ensure that the object names and the connection information in this operation do not change. This is reserved for operations such as "Export" and "Assign", which function at the object level.

Selection set

The following objects are not included in the selection set:

- Interface objects which do not contain any connection information
  Reason: A corresponding PDMS object for attribute exchange cannot be found for these objects.
- Objects which do have connection information but for which a corresponding PDMS object cannot be determined
- COMOS objects which create location objects
  Reason: Only COMOS objects which are connected are considered for inclusion in the selection set. COMOS objects that create location objects are not connected. However, the location objects created by this object are inserted.

Result

- The attribute values are exchanged in both directions as specified in the configuration. Whether the attribute value of the COMOS object is written to an attribute of the corresponding PDMS object or vice versa is also set in the configuration.
- If the user has no write rights for a corresponding PDMS object, only the attributes at the COMOS object are updated.
- Once the data exchange is complete, the status of the objects in the selection set is checked. Here only attribute values that were either transferred from COMOS to PDMS or vice versa are considered. See also chapter "Check status" operation (Page 89).

4.6.3 "Custom Refresh" operation

Purpose of the "Custom Refresh" operation

User-defined attribute values are exchanged with the "Custom Refresh" operation.

In the "COMOS Custom Refresh" window you define which attributes are to be refreshed. Information regarding the configuration of the "COMOS Custom Refresh" window: See also chapter "COMOS Custom Refresh" PDMS window (Page 115).

Call

To call the "Custom refresh" interface operation, select the "PDMS > Custom refresh" command from the context menu of an object in the Navigator.

Selection set

The operation can only be performed for a single COMOS object. In other words, the selection set only contains one object.

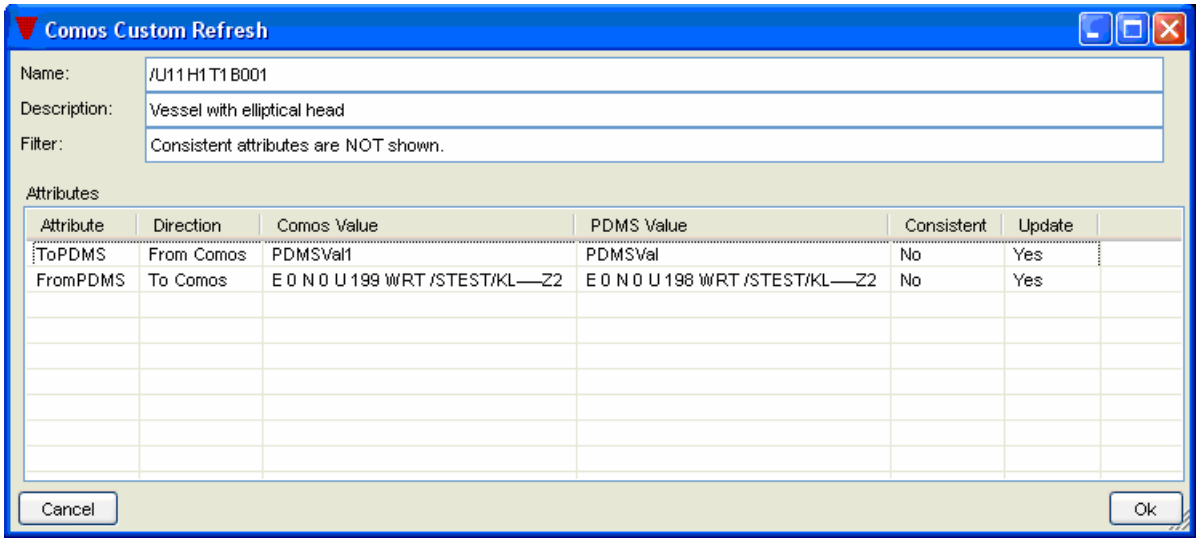
Configuring the "COMOS Custom Refresh" window

When you call the "Custom Refresh" operation in COMOS, the interface relinquishes control to PDMS. The "COMOS Custom Refresh" window then opens automatically:

Selecting the attributes for the refresh

To define which object attributes are updated, proceed as follows:

1. Select an attribute in the "Attribute" column.
2. Right-click in the list line of the required attribute to open the context menu.



   – Command "Update > Yes": The attribute is updated.
   – Command "Update > No": The attribute is not updated.
3. Repeat step 2 for all attributes whose update settings you want to define.
4. Click "OK".

Result

The attributes of the PDMS object and of the COMOS object are updated according to your settings.

4.6.4 "Assign" operation

Purpose of the "Assign" operation

You use the "Assign" operation to connect two objects that already exist in COMOS and PDMS. The operation sets the connection information at both objects and matches the names.

The operation is available in two variants:

- "PDMS > Assign> Match Names"
- "PDMS > Assign> Selected object"

The two variants differ according to the way in which a PDMS object is identified by a given COMOS object.

Requirement

In order to connect objects after identification of the PDMS object, the following conditions must be met:

- There must not yet be any connection information available at the PDMS object.
- The element type specified at the COMOS object must match the element type of the PDMS object.
- The owner restrictions must be met.
- The user who called the operation must have write rights for the PDMS object in PDMS.

Call

To call the "Assign" interface operation, select the "PDMS > Assign > ..." command from the context menu of an object in the Navigator.

Result

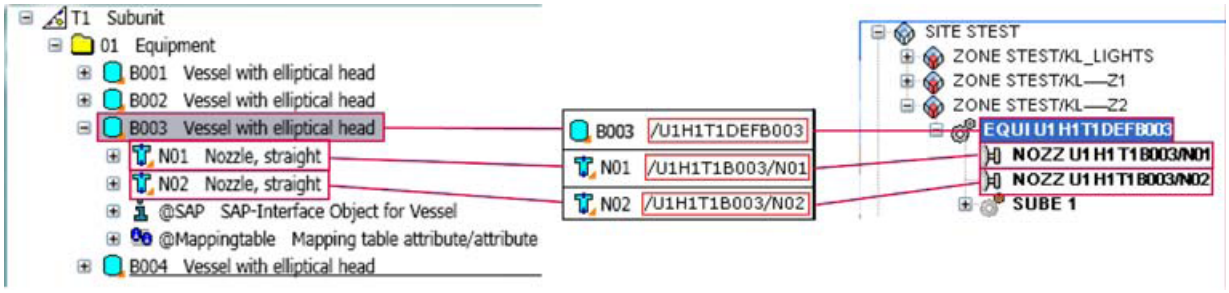
- The PDMS objects that are found are connected to the corresponding COMOS objects.
- Data exchange is performed at the attribute level for the objects connected.
- Afterwards, the attributes of interconnected objects that are relevant for a status check according to the configuration are compared.

"Assign> Match Names"

When the "Assign > Match Names" command is called, the connection to the PDMS object is established via the name generated by the subclass of the COMOS object.

If the interface finds an object with this name in PDMS, it tries to connect this PDMS object to the COMOS object. As the names have already been matched, the interface simply sets the connection information.

Example:



The name that is created via the subclass can contain name parts that are determined in PDMS. Depending on where this name part is taken from, many different names can be generated from this mask.

Two cases are differentiated in this context:

- Case 1: The COMOS object does not create location objects.
  Details: See below.
- Case 2: The COMOS object creates location objects.
  Details: See below.

"Assign> Match Names" without location objects

If the COMOS object at which the "Assign > Match Names" command was called does not create any location objects, the assignment via the generated name is not unique.

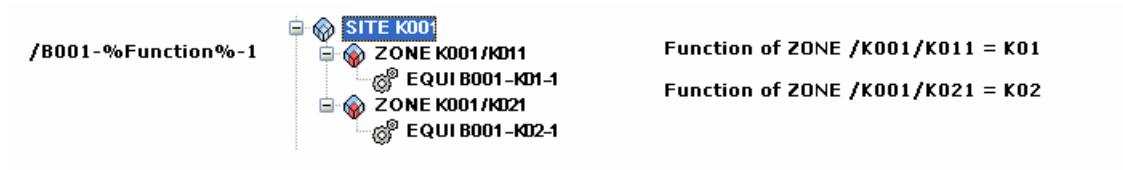
Consequence: The COMOS object is connected to the first PDMS object whose name matches the evaluated mask. This does not necessarily need to be the expected PDMS object.

The configuration is wrong at this point and must be corrected.

Example:

A COMOS object is configured in such a way that the name mask between the two name parts "B001-" and "-1" uses the value of the "Function" attribute of the owner: "/B001-%Function%-1"

In the case described in the figure below, "%Function%" is replaced with the value of the Function attribute of the zone. As a result, the assignment is not unique. Both "/B001-K01-1" and "/B001-K02-1" fulfill the condition:



"Assign> Match Names" with location objects

If the COMOS object for which the "Assign > Match Names" command was called creates location objects, the following applies:

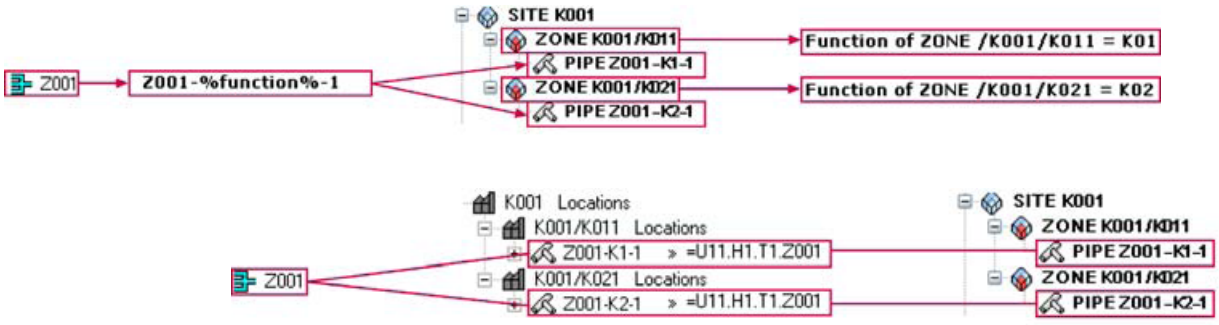
- All PDMS objects that are found are connected to COMOS objects.
- These COMOS objects can be found in the Navigator on the "Locations" tab.

- If they do not yet exist, they are created.
- You are given a reference to the object on the "Units" tab for which you called the operation.

Example:

Pipes are configured in such a way that they create location objects and have the "Z001" and "/Z001-%function%-1" masks. This mask is sufficient in PDMS for pipes "/Z001-K1-1" and "/Z001-K2-1".

Two location objects are created when the "Assign > Match Names" command is called. The location objects are connected to the corresponding PDMS objects and each location object has a reference to the "Z001" pipe:



This is behavior intentional. For example, the name mask can be generated in such a way that a name part comes from the object underneath which the object to be found is located. This makes it possible to separate a pipe that passes through various zones in PDMS into multiple parts in COMOS.

For objects which create location objects, the following applies:

- If the operation was called in the unit view, the object is added to the selection set.
- If the operation was called in the location view, the name is not generated via the subclass; instead, the name of the COMOS object matches the name of the corresponding PDMS object.

Note

If the names of location objects and of the PDMS objects which are connected to them were only manipulated using interface operations, the names will match.

"Assign> Selected object"

You use the "Assign > Selected object" command to call only selection sets with an element. The command connects the COMOS object at which you called the operation to the current element in PDMS.

If the name of the current element does not match the name generated by the subclass of the COMOS object, you can accept a name change for the PDMS object or cancel the process entirely.

4.6.5 "Unassign" operation

Purpose of the "Unassign" operation

You use the "Unassign" operation to release the connection between COMOS objects in the selection set and the connected PDMS objects.

This assumes that the COMOS objects in the selection set are exclusively those whose connection information is defined. Furthermore, this means that for COMOS objects that create location objects, the location objects are added to the selection set.

Call

To call the "Unassign" interface operation, select the "PDMS > Assign > Unassign" command from the context menu of an object in the Navigator.

Result

For all objects in the selection set, the connection information is deleted regardless of whether the counterpart is found in PDMS or not.

If the PDMS counterpart is found, the connection information at the PDMS object is also deleted.

4.6.6 "Check status" operation

Purpose of the "Check status" operation

When you select the "Check status" operation, the status of the objects from the selection set on the object level and attribute level is recalculated:

- At object level:
  A check is carried out to ascertain whether a connected PDMS object can be found for a given COMOS object.
- At attribute level:
  All attribute values designated for data matching via the configuration are compared in COMOS and PDMS.

The result of the status comparison is displayed in color in COMOS and in PDMS.

Call

To call the "Check status" interface operation, select the "PDMS > Check status" command from the context menu of an object in the Navigator.

Result

- The status of the objects from the selection set is recalculated at the object level. The following results are possible:
  - The connection information is missing at the COMOS object:
    Status = "Not available in PDMS"
  - Connection information exists at the COMOS object, but no PDMS object can be identified via the connection information:
    Status = "Deleted in PDMS"
- Finally, the correctly connected objects are recalculated on the attribute level.
- During data matching at attribute level, all attribute values designated for data matching via the configuration are compared in COMOS and PDMS. The following results are possible:
  - One or a number of attribute pairs are found with different values:
    Status = "Inconsistent"
  - All to be compared attribute values match:
    Status = "OK"
- Once the status has been calculated for all COMOS objects, the calculated status is set in COMOS for each object.
- The status for the corresponding PDMS objects that were found is set to "UDA :ComosStatus".
- The status display in COMOS is activated.

The status display in COMOS

If the status display is activated in COMOS, the objects are displayed in COMOS with their color indicating their status.

The same color display is also available on the P&ID on which the objects are placed.

Color-coded status display

In the database, the following status display colors are set by default for the COMOS PDMS Interface:

- Blue: Status = "Not available in PDMS"
- Yellow: Status = "Deleted in PDMS"
- Red: Status = "Inconsistent"
- Green: Status = "OK"

To change the color display, follow these steps:

1. Right-click on an empty location in the Navigator.
2. Select the "Legend" command from the context menu.
   Result: The "Color settings" window opens. It shows the color assignment for the currently active status display.

3. On the "Color settings" tab, specify which status value is identified by which color.
4. Confirm your entries.

Activating and deactivating the status display

To deactivate or reactivate the status display, proceed as follows:

1. Right-click on an empty location in the Navigator.
2. Select the "Status display > Switch off" command from the context menu.

To reactivate the status display, proceed as follows:

1. Right-click on an empty location in the Navigator.
2. Select the "Status display > 18 COMOS <- XML -> PDMS > Complete" or "For current user" command from the context menu.

Calling detailed status information

To provide you with detailed information on the status of the objects, a query that describes the status of the objects in greater detail has been prepared in the database. Proceed as follows to open this query:

1. Right-click on an empty location in the Navigator.
2. Select the "Status display > 18 COMOS <- XML -> PDMS > Status per XML file" command from the context menu.
   Result: The query opens. It contains all objects that were checked in the last cycle.
   The query has the following columns:
   – "Object" column: The name of the checked object
   – "Status value" column: The status value of the object
   – "Description" column: The reasons that led to the status value

4.6.7 "Rerun documents" operation

Purpose of the "Rerun documents" operation

The "Rerun documents" operation is based on the "Import DocLinks" operation. The "Import DocLinks" operation is called in PDMS. It creates DocObjs for draft objects under a COMOS document specified in the draft structure.

If the "Import DocLinks" operation is called in PDMS, it is not always ensured that the specified document already exists in COMOS. In this case the XML file remains in the documents folder which was entered in the project properties. Otherwise, it is moved to the "Imported" subfolder.

"Rerun documents" processes all in the documents folder remaining XML files again.

If you were not able to perform the "Import DocLinks" operation because the corresponding document did not yet exist in COMOS, you can create the document in COMOS. You import the missing DocLinks by calling "Rerun document import".

Call

To call the "Rerun document import" operation, select the "PDMS > Documents > Rerun import" command from the context menu of an object in the Navigator.

Result

The interface tries to localize the corresponding document for all XML files:

- The document is found: The DocLinks are created and the file is moved to the "Imported" folder.
- The document does not exist: The XML file remains in the documents folder.

See also

Exporting DocLinks (Page 101)

4.6.8 3D view operations

Call

To call a 3DView operation, select the required operation from the following context menu submenu: "PDMS > 3D view > ..."

Here you will find the following operations:

- "Add" operation (Page 92)
- "Remove" operation (Page 93)
- "Mark" operation (Page 93)
- "Zoom" operation (Page 94)
- "Select" operation (Page 95)

4.6.9 "Add" operation

Purpose of the "Add" operation

You can use the "Add" operation to add objects to the PDMS Drawlist. This corresponds to the "Add" functionality in PDMS.

Selection set

Only objects with connection information are added to the selection set. If a COMOS object creates location objects, the object itself is not added to the selection set; only its location objects are added.

Result

For all objects in the selection set, the operation checks whether a connected PDMS object exists. If the operation finds the PDMS object, it is added to the Drawlist in PDMS.

4.6.10 "Remove" operation

Purpose of the "Remove" operation

You can use the "Remove" operation to remove objects from the PDMS Drawlist. It corresponds to the "Remove" functionality in PDMS.

Result

For all objects in the selection set, the operation checks whether a connected PDMS object exists. If the PDMS object is found, it is removed from the Drawlist in PDMS.

4.6.11 "Mark" operation

Purpose of the "Mark" operation

You use the "Mark" interface operation to mark objects in PDMS. It corresponds to the "Mark" functionality in PDMS.

Selection set

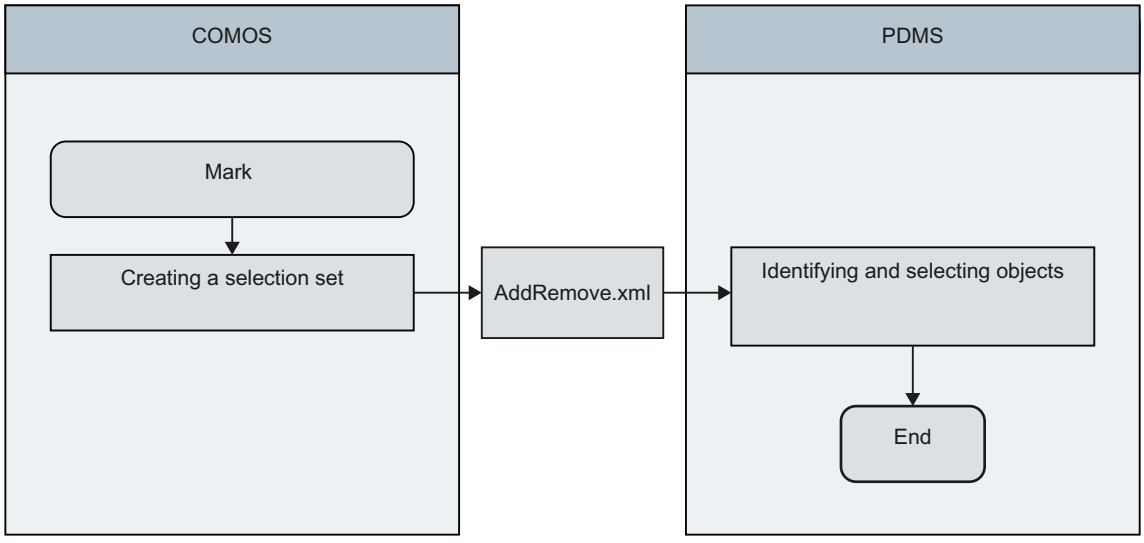
Only objects with connection information are added to the selection set. If a COMOS object creates location objects, the object itself is not added to the selection set; only its location objects are added.

Result

For all objects in the selection set, the operation checks whether a connected PDMS object exists. If the PDMS object is found, it is marked in PDMS.

Flow diagram

The following diagram illustrates the sequence of the "Mark" interface operation:



4.6.12 "Zoom" operation

Purpose of the "Zoom" operation

You use the "Zoom" interface operation to zoom to objects in PDMS.

Selection set

Only objects with connection information are added to the selection set. If a COMOS object creates location objects, the object itself is not added to the selection set; only its location objects are added.

Result

For all objects in the selection set, the operation checks whether a connected PDMS object exists.

If the PDMS object is found, it is added to the Drawlist in PDMS. Then, the viewport in PDMS is set up so that the objects are visible and zooming is applied this area.

If several objects are found, the 3D Viewport is oriented in such a way that all the objects from the selection quantity are visible together.

4.6.13 "Select" operation

Purpose of the "Select" operation

You have multiple options for selecting a PDMS object from COMOS.

The variants differ in the manner in which the object to be selected is identified in PDMS.

- "... > Select > Connected object"
  If a PDMS object with corresponding connection information exists and the name matches, it is selected.
- "... > Select > By RefNo"
  If a PDMS object with corresponding connection information exists, it is selected.
- "... > Select > By Name"
  If a PDMS object with the same name is found, it is selected.

Call

To call the "Select" interface operation, in the context menu, select a variant of the operation from the "PDMS > 3D View > Select" submenu.

Selection set

The operations are only available for selection sets that contain a single element.

Result

When the corresponding PDMS object for a COMOS object is found, it is selected in PDMS. In other words, the object becomes the current element in PDMS.

4.7 PDMS-side interface operations

4.7.1 Operations on the current element

The following interface operations are available for the current element:

- "Navigate"
- "Unassign"
- "Reassign"

4.7.2 "Navigate" operation

Purpose of the "Navigate" operation

You can use the "Navigate" operation to navigate from the current element in PDMS to the object connected in COMOS.

Requirement

The current element is connected to a COMOS object.

Call

To call the "Navigate" operation, select the "COMOS > CE > Navigate" command from the PDMS menu bar.

Note

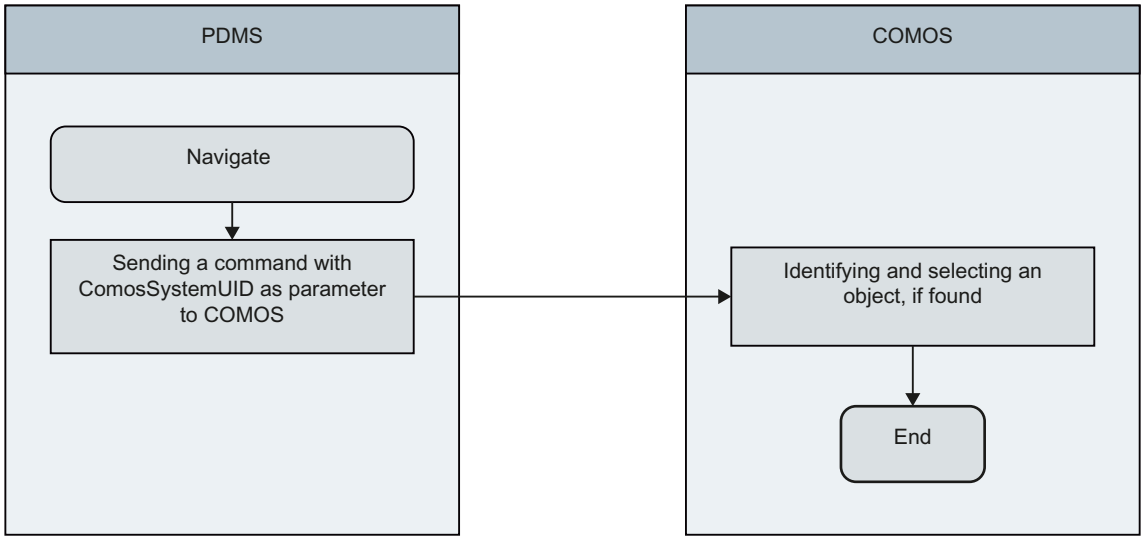
Complete connection information is not mandatory

The COMOS object is identified only via the system UID. During this operation, other connection information such as Name or RefNo is not considered.

When an object is found in COMOS, this does not necessarily mean that it is fully connected to the current element.

Flow diagram

The following diagram illustrates the sequence of the "Navigate" interface operation:



4.7.3 "Unassign" operation at the CE

Purpose of the "Unassign" operation

You can use the "Unassign" operation to release the connection between the current element and its connected COMOS object.

If the current element is potentially connected, its connection information is deleted regardless of whether or not a corresponding object is found in COMOS.

Call

To call the "Unassign" operation, select the "COMOS > CE > Unassign" command from the PDMS menu bar.

4.7.4 "Reassign" operation

Purpose of the "Reassign" operation

The "Reassign" operation updates the location pointer of the connected COMOS object.

Requirement

The current element must be connected to a COMOS object which has a reference to a location in the COMOS location tree.

Call

To call the "Reassign" operation, select the "COMOS > CE > Reassign" command from the PDMS menu bar.

Application example

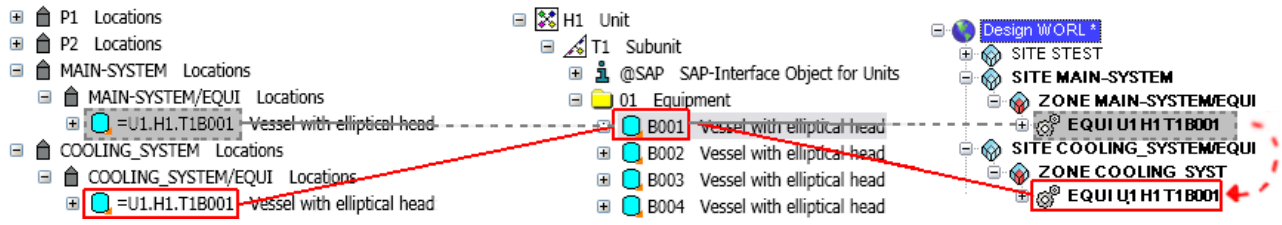
The COMOS object "A10 > A10 >A10 > B001" is connected to the PDMS object "/A10A10A10B001".

The configuration of the subclasses specifies that the COMOS object has a reference to a location which corresponds to the SITE-ZONE structure in PDMS.

Once you have moved the PDMS object to a different ZONE in PDMS, call the "Reassign" operation.

Result: The reference of the COMOS object is updated so that it corresponds to the new SITE-ZONE structure.

The following figure illustrates this process:



4.7.5 Operations that are called in the "Export to COMOS" window

In the "Export to COMOS" window you can call the following interface operations for objects in the result list:

- "Export"
  See also chapter "Export" operation (Page 98).
- "Unassign"
  See also chapter The "Unassign" operation in the "Export to COMOS" window (Page 99).
- "Reassign"
- "Check status"
  See also chapter "Check status" operation (Page 100).

The following commands are also available:

- "Select all"
- "Navigate in PDMS"
- "Navigate in COMOS"
- "Add to Drawlist"

See also

"Query COMOS data" operation (Page 100)

Exporting DocLinks (Page 101)

4.7.6 "Export" operation

Purpose of the "Export" operation

The "Export" operation in PDMS is the same as the "Export" operation in COMOS.

If COMOS objects corresponding to PDMS objects are missing, these are created and connected.

Call

1. Select the following menu in PDMS: "COMOS > Interface > Export to CE".
2. Configure the "Export to COMOS" window so that a result list containing PDMS objects is displayed in the "PDMS-Objects" area. Information regarding configuration: see the link below.
3. Select the objects you want to export, then select the "Export" command from the context menu.

Result

- The corresponding COMOS objects are created.
  In the configuration you can define where in COMOS the objects are created.
- The objects that were created in COMOS are connected to the corresponding PDMS objects. The objects are updated at the attribute level.
- If an object already exists in COMOS, it is not connected. No data exchange takes place either.

See also

Configuring the "Export to COMOS" window (Page 77)

4.7.7 The "Unassign" operation in the "Export to COMOS" window

Purpose of the "Unassign" operation

The operation deletes the connection information for all objects in the selection set regardless of whether a corresponding object is found in COMOS.

It has the same effect as the operation which is called in the PDMS menu by selecting the "COMOS > CE > Unassign" command. The only difference is that the operation is applied to all objects in the selection set.

Call

1. Select the following menu in PDMS: "COMOS > Interface > Export to CE"
2. Configure the "Export to COMOS" window so that a result list containing PDMS objects is displayed in the "PDMS-Objects" area. Information regarding configuration: see the link below.
3. Select the objects whose connection information you want to delete, then select the "Unassign" command from the context menu.

See also

Configuring the "Export to COMOS" window (Page 77)

4.7.8 "Check status" operation

Purpose of the "Check status" operation

The operation is called for a selection set of objects in PDMS. It allows you to identify PDMS objects to which no COMOS object has been assigned.

The operation checks whether the objects in the selection set have connection information and performs a data matching at the attribute level for the connected objects.

Call

1. Select the following menu in PDMS: "COMOS > Interface > Export to CE"
2. Configure the "Export to COMOS" window so that a result list containing PDMS objects is displayed in the "PDMS-Objects" area.
3. Select the objects whose status you wish to check, then select the "Check Status" command from the context menu.

See also

"Check status" operation (Page 89)

Configuring the "Export to COMOS" window (Page 77)

4.7.9 "Query COMOS data" operation

Purpose of the "Query COMOS data" operation

From within PDMS, you can configure a query that was predefined in COMOS, start it, and view the result in PDMS.

Call

1. Select the "COMOS > Interface > Query COMOS data" command from the PDMS menu bar.
   Result: The "Query COMOS" window opens.
2. Configure the following tabs in the window:
   - "Query"
   - "Filter"
   - "Execute"
3. Go to the "Execute" tab and click the "Run Query" button.
   The "Query COMOS data" operation is executed.

Result

You can view the following:

- Results of the query in Excel or in a web browser.
- A preview of the results of the query in the "Query COMOS" window.
- Save the results of the query in a file.
- Should errors occur during execution of the query: Error log in the "Query COMOS" window.

See also

Configuring the "Query COMOS" window (Page 78)

4.7.10 Exporting DocLinks

You export DocObjects for Draft objects with an DDNM attribute in order to create DocObjects in COMOS.

DDNM = attribute at a PDMS Draft Label

Requirement

- The DDNM attribute references a design object which is connected to a COMOS object.
- Your administrator has defined an attribute or a UDA at one or a number of objects in the draft structure in which the name or the system UID of the document for which DocObjs are to be created is entered.

Preparing the export

1. Open the "Draft" module in PDMS.
2. In the draft structure, select one of the nodes at which the attribute or the UDA controlled by the "Import DocLinks" operation is available.
3. Enter one of the following values in the attribute or the UDA at the node:
   - The name of the COMOS document for which you want to import DocObjs.
   - The system UID of the COMOS document for which you want to import DocObjs. The search process speeds up when you enter the system UID.
4. Select the following menu in PDMS: "COMOS > Interface > Export to COMOS"
   Result:
   - The "Export to COMOS" window opens.
   - The structure of the window is virtually identical to the "Export to COMOS" window in the Design module.
5. Configure the "Export to COMOS" window as described below.

6. In the "PDMS-Objects" area, select the draft objects for which you want to create DocLinks.
7. Select the "Export DocLinks" command from the context menu.

Configuring the "Export to COMOS" window and exporting DocLinks

1. "Select" area:
   - "Dept.", "Regi.", Drwg." fields:
     If the "Track CE" option is activated: Set automatically, depending on which node was selected when the window was called in the draft structure.
   - "Object Type" list:
     Select one of the subclasses of the here listed "Document" class. The "Import DocLinks" then uses the for the subclass specified settings.
2. "Standard Filter" and "Filter Assigned" areas:
   Define filters for the objects in the result list. Information regarding this: see the link below.
3. "Select" area: Click the "Find" button.
   Result:
   All draft objects located below the selected node and meeting the criteria you defined are displayed in the "PDMS Objects" area.
   Depending on how the subclass you selected is configured, the following objects may be filtered out of the result list:
   - Objects without names.
   - Objects whose owners do not comply with the owner restriction rules defined in the subclass.
4. Optional: "PDMS Objects" area: Define filters for attribute values in the result list.
5. In the "PDMS Objects" area, select the objects for which you want to create DocObjs.
6. Select the "Export DocLinks" command from the context menu.

Result

- The interface uses the subclass you selected to determine the name or the system UID of the COMOS document for which DocObjs are created.
- If the subclass defines filters, these are also taken into account when searching for the document.
- The operation is aborted in the following cases:
  - If the interface finds more than one document with this name.
  - If the interface does not find any documents with this name.

  Consequence:
  - The XML file with the selection set remains in the documents folder which was specified in the project properties in the "Document directory" field of the "PDMS/E3D Engineering Interface" tab.
  - You can create the DocObjs within COMOS at a later point in time.

- Once the interface has found the document, DocObjs are created for the objects in the selection set.
- The XML file is moved from the documents folder to the "Imported" subfolder.

See also

Working in the "Export to COMOS" window (Page 75)

Importing technical specifications of PDMS 5

5.1 General

You can export all technical specifications (SPEC type) for which catref points to a SCOM from PDMS and import them into COMOS.

Data exchange between COMOS and PDMS takes place offline by means of XML files.

Note

Use pipe spec import only within one working layer

Use pipe class import only within one working layer. Because the release mechanism of working layers does not support pipe spec import, you may experience data loss if you work with several working layers.

5.2 Reading pipe specs from an XML file

Requirement

You have exported a pipe spec in PDMS using the main menu command "COMOS > Export > CE" and stored it in an XML file.

Procedure

1. In COMOS, select the "PDMS/E3D pipe spec import" command from the menu "Plugins > CAE Apps > AVEVA".
2. Click the "Load default settings" button.
   The default settings are loaded.

Field	Description
"Base object for catalog folder"	Group object for the imported catalogs and base object for the generated catalog objects. Default: "@30 > M27 > A10 > A50 Folders for catalogs"
"Base object for catalog component"	Base object for the generated catalog components. Default: "@30 > M27 > A10 > A60 PDMS Cate Object"
"Base object for pipe spec folder"	Base object for the imported pipe specs. Default: "@50 > A10 > A10 > B10 Pipe spec catalog"
"Base object for imported pipe specs"	Base object for the generated pipe spec objects. Default: "@20 > C90 > A10 > A20 > A50 Template object for PDMS pipe spec"
"Base object for pipe spec elements"	Base object for pipe spec elements. Default: "@30 > M27 > A10 > A70 PDMS SPCO Object"

3. Click the "Import Specifications" button.
4. Select the XML file that you have saved previously.
5. Confirm your entries.
6. Optional: Save the log file using the following button:

Result

The pipe spec is imported from the XML file. The log information is displayed in the bottom pane of the window.

- The pipe spec catalog is created under the object that you specified in the "Base object for catalog folder" field.
- The pipe spec is created under the object that you specified in the "Base object for pipe spec folder" field. If the XML file also includes design templates, these are also created at this position.

COMOS Design Viewer 6

6.1 Opening the COMOS PDMS/E3D engineering interface in the COMOS Design Viewer

Terminology: COMOS Design View = Construction Assistant

You can open the COMOS PDMS/E3D Engineering Interface in COMOS Design View mode. COMOS is opened write-protected in the process. If a write operation accesses COMOS, the write protection is revoked until the operation is complete. The user cannot modify any data manually in COMOS.

Requirement

See chapter Opening the COMOS PDMS/E3D Engineering Interface (Page 27).

Procedure

1. Open COMOS in Design View mode. See also chapter COMOS Design Viewer (Page 107).
2. Click the "PDMS/E3D Engineering Interface" entry in the "Plugins > Basic" menu in COMOS.
3. Switch to PDMS/E3D.
4. Call the !!SAGLaunchInterface() function in the command line.

Result

- The connection between COMOS and PDMS/E3D is established.
- The engineering interface screen opens in PDMS/E3D.
- The "MessageSystem <communication mode>" window opens in COMOS.

The COMOS Design Viewer is an independent application that offers a limited number of COMOS standard functions as well as all the functions that are necessary for synchronizing your P&ID and 3D data in COMOS and PDMS.

With the exception of data exchange operations, the COMOS Design Viewer is write-protected. This means:

- You can create or change objects, reports or queries.
  You can only create or change objects by executing an interface operation.
- The project configuration must be carried out in COMOS.
  This pertains to, among other things, the following objects:
  - "Folders for class/subclass definitions" object
  - Subclass definition objects
  - Base objects, engineering objects, selection lists, and so on

6.2 Starting the COMOS Design Viewer

Procedure

1. Click "Start > Run" in the Windows task bar.
2. In the "Run" window, enter the path to your Comos.exe file and the /MD:CA command in the input line, separated by a space:
   Example: C:\Comos\PT001_Vega\Bin\Comos.exe /MD:CA

Result

The COMOS Design Viewer opens.

Changing the database or project

To open or change a database or to switch projects or working layers, click the "Open project" button which is displayed on the menu bar of the COMOS Design Viewer.

6.3 The "Documents" area

Purpose

The "Documents" area is a separate window which displays the contents of a COMOS report, the properties of an object, or the contents of a query.

Opening the "Documents" area

The window opens when you open a project.

Content of the "Documents" area

It is possible to display a report, the properties, and a query at the same time: Their contents are displayed on tabs so that you can switch between the contents.

However, it is not possible to simultaneously display more than one report, the properties of multiple objects, or multiple queries.

6.4 Status information

Show status

You can display status information for the COMOS objects in the COMOS Design Viewer.

Procedure

To activate the status display in the Navigator, proceed as follows:

1. Right-click on an empty location in the Navigator.
2. Select the "Status display > 18 COMOS <- XML -> PDMS > Complete" command from the context menu.

Result

The status display is activated and color-codes the objects according to the default settings.

6.5 Context menus

You control the interface mainly using the context menu, which is available to you in the Navigator, in P&IDs and in queries.

The context menu contains various commands:

- General commands as in the full version of COMOS
  These include "Navigate", "Properties".
- Interface-specific commands

Interface-specific commands

You can find the following interface-specific commands in the "PDMS > ..." context menu:

- "... > Options > Select automatically":
  If selected: When you select and object in an open report, it is automatically selected in the Navigator.
- Commands which start interface operations for data exchange.
  Detailed information about interface operations: see the link below.

See also

Interface operations in COMOS (Page 80)

Material management

7

Material management is used to create lists of estimates and purchasing lists in COMOS. COMOS PDMS Integration provides all COMOS material management functions.

With the help of MTO options from Aveva PDMS, you can transfer material to COMOS and have MTO objects created from the material for the material management. For this purpose, all the required data from Aveva PDMS is written to an XML file, which you import into COMOS.

You will find more detailed information on this topic in the "Material Management" manual.

COMOS 3D viewing

8

8.1 Reference

You can find documentation on this topic in the Installation Package of the "COMOS Walkinside" software. You can find additional information on this topic in the "COMOS Walkinside" manual, keyword "COMOS Walkinside Integration".

References 9

9.1 COMOS PDMS Integration

9.1.1 User interface reference

9.1.1.1 "COMOS Custom Refresh" PDMS window

Control element	Description
"Name" field	The name of the PDMS object. The object is connected to the COMOS object that you called using the "Custom Refresh" operation. The field is automatically set.
"Description" field	The description of the PDMS object. The field is automatically set.
"Filter" field	The field evaluates whether the "Attributes" list displays attributes whose values differ in COMOS and PDMS. The field is automatically set.
"Attributes" list	Default setting: List of the attributes which differ in COMOS and PDMS You can also display the consistent attributes: • Select any row in the "Attributes" list. • Select the "Display consistent" command from the context menu.

9.1.1.2 "Export to COMOS" PDMS window

The "Export to COMOS" window has the following control areas:

- "Select"
- "Standard Filter"
- "Filter Assigned"
- "PDMS Objects"

"Select" control group

In this control group, you specify the basic settings for filtering the result list.

Control element	Description
"Site" field	Displays the site currently selected in PDMS. If the "Track CE" option is deactivated: Enter the site below which the objects are searched and for which you want to execute an interface operation.
"Zone" field	Displays the zone currently selected in PDMS. If the "Track CE" option is deactivated: Enter the zone below which the objects are searched and for which you want to execute an interface operation.
"Object Type" list	Contains all available subclasses. Select a subclass to define the subclass to which the objects in the selection set belong.
"Track CE" option	Default setting: Activated When you activate this option, the "Site" and "Zone" fields are set automatically depending on the Current Element in the Design Navigator.
"Auto Find" option	When you activate this option, the result list is refreshed automatically once a filter is set.
"Find" button	Refreshes the result list.

"Standard Filter" control group

In this control group, you can choose to filter the result list on the basis of specific texts:

Control element	Description
"Name" field	Here you define a name mask. The objects in the result list are filtered on the basis of the name mask. Enter a specific name or replace name parts with placeholders: • "?" stands for a letter • "*" stands for multiple letters Examples: • "BO*" would display all objects whose names begin with BO and meet the remaining filter criteria. • "*1?N0*" would display all objects whose names contain the letter group "1 + any letter + N0" (e.g. B001/N01) and meet the remaining filter criteria.
"Description" field	Similar to the "Name" field but applied to the description of the objects.
"Display Text" field	In a similar way to the name filter, here you can filter a row in the list using the entire display text.

"Filter Assigned" control group

In this control group, you can choose to filter the result list on the basis of connection information:

Control element	Description
"Assigned" option	Activated: Only objects which are potentially connected are displayed in the list. In the case of potentially connected objects, the "Assigned" column has the value "TRUE".
"Unassigned" option	Activated: Only objects which are not connected are displayed in the result list. For these objects, the "Assigned" column has the value "FALSE".
"Both" option	Activated: The connection information is not taken into account during filtering.

"PDMS Objects" control group

The result list is displayed in this control group. The result list is structured as follows:

Column name	Function
"Name"	The name of the object in PDMS
"Description"	The description of the object in PDMS
"Assign"	The connection information Possible values: "TRUE" or "FALSE"
Attribute columns	Which columns are displayed here depends on the configuration of your administrator: Your administrator can define attributes at the subclass definition objects in COMOS which can be used for filtering during data exchange. A column is displayed to the right of the "Assigned" column for each attribute defined by your administrator in COMOS. The columns read out the attribute value of the object.

9.1.1.3 "Query COMOS" PDMS window

"Query" tab

On this tab, select the COMOS query you wish to execute.

Control element	Description
"Query" list	In this list, you find all queries that were defined in advance by your administrator in the form of subclass definition objects. Select a query. Result: The remaining control elements of the "Query" tab are filled with values from the subclass.
"Description" field	The description text of the selected query.
"Start Object" field	The start object of the query defined in the COMOS query.
"Has start query" option	Indicates whether or not a start query was predefined.

Control element	Description
"Output format" field	The file format of the result file.
"Delimiter" field	If the output format is CSV: The character separating the data in the result file.

"Filters" tab

The "Filters" tab is where you configure filters for the query. Only objects that meet all filter conditions are included in the result file.

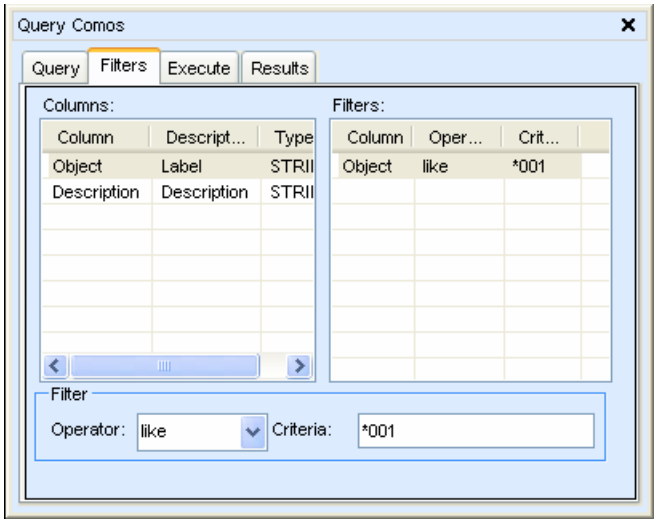
Control element	Description
"Columns" area	Lists all columns that were defined for the COMOS query. Each row corresponds to one column. The following column data is displayed: • "Columns" column: The column name • "Label" column: The description text of the column in the query • "Type" column: The data type of the column value
"Filters" area	This area shows which filters you have defined for the columns of the query. It is possible to define multiple filters for a column.

Operators on the "Filters" tab

The following table describes which operators are available for selection and how they interact with the criterion entered in "Criteria":

Operator	Criterion
"eq"	The cell value must match the value specified in the "Criteria" field.
"neq"	The cell value must not match the value in the "Criteria" field.
"gt"	The first character of the cell value that is different from the corresponding value in the "Criteria" field must be greater than the corresponding character in the "Criteria" field.
"lt"	The first character of the cell value that is different from the corresponding value in the "Criteria" field must be smaller than the corresponding character in the "Criteria" field.
"like"	The cell value must correspond to a mask specified in the "Criteria" field. In this mask, the following characters are used as placeholders: • "?" • "*"

Example of a filter on the "Filters" tab



The filter set in the figure above returns "true" for all rows that meet the following condition:

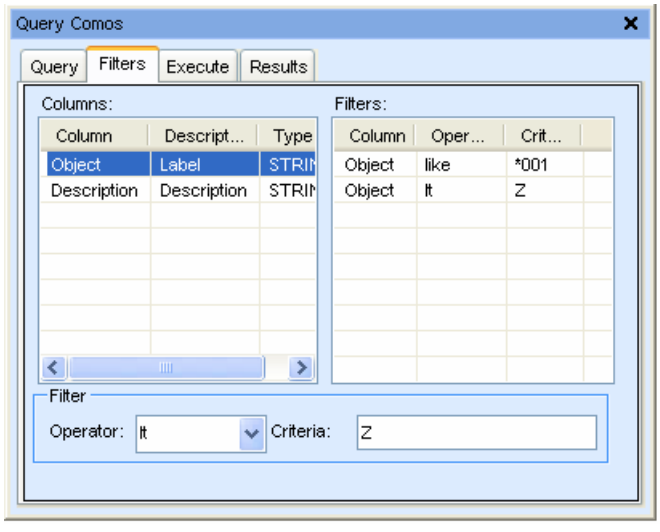
Object like "*001"

Thus, the row with the value "B001" in the "Object" column will be inserted in the result file, but not the row with the value "B002".

If you also set the filter Object lt "Z", the row with the value "B001" in the "Object" column will be inserted into the result file, but not the row with the value "Z001".

The following applies:

- "Z001" like "*001" = True
  "Z001" lt "Z" = False
- "B001" like "*001" = True
  "B001" lt "Z" = True
  "B001" is not sufficient for the "Z001" filters.



"Execute" tab

The "Execute" tab is where you define the start object of the query.

If you do not define a start object, the start object predefined by your administrator in the subclass definition object is used.

"Start Object" area

Option	Description
"System UID"	The start object is identified on the basis of its system UID.
"Name"	The start object is identified on the basis of its name.
"Label"	The start object is identified on the basis of its description.
"System full name"	The start object is identified on the basis of its SystemFullName.
"Start query"	The start object is selected from a range of objects which are identified by means of a start object query.

"Start Object" field

Control element	Description
"Start object" field	Depending on which option you set in the "Start object" area, here you enter the system UID, the name, etc. If you selected the "Start query" option: If your administrator has prepared a start object query, this is executed automatically by activating the option. The start object query returns a range of possible start objects, which you can select in the "Start Object" list.

"Results" tab

This tab shows the results of a query.

Control element	Description
"Number of rows" field	Indicates the number of rows in the result file.
"Error code" field	Indicates whether the operation was performed successfully or not. Value = "0": The query has been executed successfully.
"Error Text" field	Describes an error that has occurred. Remains empty if no errors have occurred.

"Result" area

Use the buttons in this area to select how you wish to proceed with the result file for the "Query COMOS data" operation:

Symbol	
	Depending on the output format: Have the result displayed in one of the following applications: • In Microsoft Excel Output format: CSV • In a web browser Output format: XML
	Save the result in a file. The format in which the result is saved corresponds to the output format.
	View a preview of the result in the "Result" area.

9.1.1.4 COMOS Design Viewer: Upper area

The area at the top contains the following buttons:

Symbol	Meaning	Function
	"Open database"	Opens the "Open database" window.
	"Open project"	Opens the "Open database" and "Open project" windows.
	"Construction Assistant Log"	Hides/shows the log in the lower area of the COMOS Design Viewer. The COMOS Design Viewer log outputs the log text and error messages of the interface operations.
	"Print"	Opens the COMOS Print Manager.

9.1.2 List of interface operations and internal names

Names of interface operations in the software

The following table provides you with an overview of how the interface operations are named in the software.

Background: Whenever you implement a pre/post executable, you transfer the name of the interface operation before or after the function is executed in the `InterfaceOp` parameter.

Interface operations that start in COMOS

Name in documentation	Name in software
"Export"	"Export"
"Refresh"	"Refresh"
"Custom Refresh"	"CustomRefresh"
"Assign"	"Assign"

Name in documentation	Name in software
"Unassign"	"Unassign"
"Check status"	"CheckStatus"
"Rerun documents"	"RerunDocuments"
"Add"	"Add"
"Remove"	"Remove"
"Mark"	"Mark"
"Zoom"	"Zoom"
"Select"	"Select"

Interface operations that start in PDMS

Name in documentation	Name in software
"Import"	"Import"
"Navigate"	"Navigate"
"Unassign"	"Unassign"
"Reassign"	"Reassign"
"Check status"	"CheckStatus"
"Query COMOS data"	"QueryComosData"
"Import DocLinks"	"ImportDocLinks"

You can find additional information on this topic in the "3D Integration Administration" manual, keyword "Using pre/post executables".

9.2 Menu "Extra > COMOS 3D Review"

Entry	Description
"3D Piping"	Displays 3D data in a 3D Editor. The Editor is designed for editing P&IDs based on 3D data.
"Electrical Engineering"	Displays 3D data in a 3D Editor. The Editor is designed for diagrams based on 3D data for control cabinet design.